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Two-stage robust optimization approach for enhanced community resilience under tornado hazards

Mehdi Ansari ^a, Juan S. Borrero ^b*, Andrés D. González ^c^a American Airlines, Fort Worth, TX, United States of America^b Department of Industrial and Management Systems Engineering, University of South Florida, United States of America^c School of Industrial and Systems Engineering, University of Oklahoma, United States of America

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ABSTRACT

Catastrophic tornadoes cause severe damage and are a threat to human wellbeing, making it critical to determine mitigation strategies to reduce their impact. One such strategy, following recent research, is to retrofit existing structures. To this end, in this article we propose a model that considers a decision-maker (a government agency or a public-private consortium) who seeks to allocate resources to retrofit and recover wood-frame residential structures, to minimize the population dislocation due to an uncertain tornado. In the first stage the decision-maker selects the retrofitting strategies, and in the second stage the recovery decisions are made after observing the tornado. As tornado paths cannot be forecasted reliably, we take a worst-case approach to uncertainty where paths are modeled as arbitrary line segments on the plane. Under the assumption that an area is affected if it is sufficiently close to the tornado path, the problem is framed as a two-stage robust optimization problem with a mixed-integer non-linear uncertainty set. We solve this problem by using a decomposition column-and-constraint generation algorithm that solves a two-level integer problem at each iteration. This problem, in turn, is solved by a decomposition branch-and-cut method that exploits the geometry of the uncertainty set. To illustrate the model's applicability, we present a case study based on Joplin, Missouri. Our results show that there can be up to 20% reductions in worst-case population dislocation by investing \$15 million in retrofitting and recovery and that our approach outperforms other retrofitting policies.

1. Introduction

Tornadoes are vertical columns of rotating air that are spawned from supercell thunderstorms caused by rotating updrafts that form because of the shear in the environmental wind field (NOAA, 2022d). Catastrophic tornadoes are common natural disasters that happen in many populated regions around the world and are particularly concerning in dense high-risk urban areas because of their damage potential. On average, more than 1200 tornadoes happen in the US annually which cause around 60 fatalities (NOAA, 2022a). Tornadoes have caused damage costs that range between \$183 million to \$9.493 billion per year in the US within the past two decades (NOAA, 2022f). The intensity of tornadoes is measured by the Enhanced Fujita (EF) scale, which ranks a tornado from 0 (weakest) through 5 (most violent) based on an estimation of their average wind speed. The average wind speed, in turn, is estimated by comparing the observed tornado damages with a list of Damage Indicators and Degrees of Damage (NOAA, 2022b).

Fatalities and injuries from tornadoes have decreased in the past few decades thanks to warning systems based on weather forecasts (Koliou

& van de Lindt, 2020; Standohar-Alfano et al., 2017). For example, the Storm Prediction Center at the National Weather Service keeps a Day 1–8 Convective Outlook for all the US and issues watches or warnings based on the possibility or observation of tornadoes (NOAA, 2022c). Tornado warnings are typically issued as soon as a rotating supercell has been identified by a weather radar; these warnings instruct people in the affected area to take shelter immediately. Even though the warning systems have caused a great reduction in fatalities (FEMA, 2022b), their short lead times (of the order of minutes) do not allow to prevent physical structures from possible destruction. This is further aggravated by the fact that more than 80% of building stocks in the US are wood-frame buildings that are highly vulnerable to wind damage (van de Lindt & Dao, 2009).

Besides weather forecasting, other alternatives can be employed to reduce the impact of tornadoes. In the past decade, several studies have shown that retrofitting strategies with simple and inexpensive actions, e.g., improving the roof cover or enhancing the roof sheathing

* Corresponding author.

E-mail address: jsborrero@usf.edu (J.S. Borrero).<https://doi.org/10.1016/j.ejor.2025.03.001>

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nailing pattern of a house, can improve building codes to make wood-frame buildings more resistant to tornado damage, particularly from damaging tornadoes of EF2 intensity or less (Koliou & van de Lindt, 2020; Masoomi et al., 2018; Ripberger et al., 2018; Simmons et al., 2015; Wang et al., 2021). In particular, Ripberger et al. (2018) show that a 30% reduction in lifetime damage can be expected by enhancing existing wood-frame buildings with simple retrofitting actions. Similarly, recent research has studied different recovery strategies and has shown that they have the potential to significantly improve the restoration time and reduce population dislocation (Farokhnia et al., 2020; Koliou & van de Lindt, 2020; Masoomi & van de Lindt, 2018).

At the federal level in the US, the use of retrofitting has been identified as a valuable strategy to reduce the impact of natural disasters (FEMA, 2021, 2022a; The White House, 2022). At the local level, however, such strategies have not been implemented yet. In fact, there are federal programs that provide general guidelines for local administrators (e.g., the emergency management agencies of cities and towns) to design and fund retrofitting plans (FEMA, 2021). Motivated by this need, the main goal of this paper is to present a model to guide retrofitting and recovery investments in order to enhance the community resilience of an urban area subject to tornado hazards. Specifically, the primary objective of this research is to establish a robust strategic framework for guiding long-term planning and budget allocation for tornado preparedness in urban regions. Our proposed method aims to empower decision-makers by providing advanced insights into the anticipated budget requirements for each region along with the optimal allocation proportions within the assigned budget for retrofitting and repair. Given the unpredictable and infrequent nature of tornado events, planning for individual occurrences beforehand is impractical. However, our approach enables decision-makers to proactively plan and prepare for the worst-case scenario, aligning resources with strategic priorities in a manner that enhances overall community resilience.

In the context of natural and anthropogenic hazards, community resilience refers to the ability of a community to withstand, adapt to, and recover from the impact of a disruptive event. This encompasses diverse efforts to prepare, mitigate, respond, recover, and adapt social, economic, and physical infrastructure, so that the underlying community and its functions can perform adequately over time (Berkes & Ross, 2013). As such, community resilience involves a complex multidimensional perspective with profound dynamic socio-physical factors (Koliou et al., 2020).

Given the intrinsic nature of resilience-related challenges in communities, uncertainty considerations are critical. Multiple sources of uncertainty affect the way communities plan for and recover from disruptive events, including but not limited to uncertainties associated with resource availability and costs, time to complete mitigation and recovery tasks, and more importantly, the location, nature, and level of intensity of disruptive events. In this context, approaches such as robust optimization can provide key insights than other approaches based on simulation or stochastic programming cannot. Through robust optimization techniques, it is possible to determine the worst-case scenario associated with a set of possible events, to provide adequate mitigation and recovery plans (Wang & Paul, 2020). Using robust optimization approaches, communities can reduce critical vulnerabilities in their infrastructure systems, by identifying and addressing weaknesses and risks that might otherwise be overlooked (Yuan et al., 2016a).

In this work, we assume that there is a decision-maker (a government agency or a public-private consortium; for instance, the emergency management department of a city and home insurance companies) that can invest a limited budget to retrofit and recover structures across a geographical area of interest, in order to maximize the community resilience due to an uncertain tornado. We model this decision problem as a two-stage robust optimization model. The first-stage decisions are made before the realization of a tornado and determine

what structures have to be retrofitted and to what extent. The second-stage decisions determine recovery strategies for the locations that are damaged by the tornado. Without loss of generality, in this article we consider total *population dislocation*, which is the involuntary movement of people from their residential sites after a severe tornado, as a metric for community resilience to be minimized (therefore, maximizing community resilience would be equivalent to minimizing population dislocation). Our proposed formulation, however, allows for other metrics that assess community resilience.

We formulate the first- and second- stage problems as integer programming problems (IP) whose decision variables select retrofitting strategies and recovery strategies, respectively, for each location. Both stages share the same budget, which models that, at least a priori, the decision-maker does not know what proportion of the budget should be spent on retrofitting and what proportion on recovery. On the other hand, as an accurate prediction of the location/time where a tornado forms, the motion direction, and its magnitude, is out of reach of the current technology (NOAA, 2022e), we approach the uncertainty using a robust lens. That is, we assume that given any retrofitting plan, the corresponding worst-case tornado damage happens. In order to have an adequate balance between accuracy and tractability, we model tornado paths as arbitrary line segments on the plane and assume that an area is affected by the tornado if it is sufficiently close to the line segment. This approach results in a mixed-integer non-linear uncertainty set.

In order to solve the problem, we employ an exact column-and-constraint generation (C&CG) algorithm based on the method discussed by Zeng and Zhao (2013). Here, at each iteration, a master problem solves a relaxation of the original problem over a subset of possible tornado scenarios. New scenarios are iteratively generated in a subproblem and added to the master until the objective value does not change. In contrast with previous models that employ the C&CG algorithm, for instance Velasquez et al. (2020), Yuan et al. (2016b), Zeng and Zhao (2013), our subproblem is a challenging max-min non-linear integer problem that cannot be solved by standard ‘dualize and combine’ techniques. Thus, the subproblem is solved by a decomposition branch-and-cut algorithm that exploits the geometric properties of the uncertainty set. Particularly, we derive ‘double’ and ‘triple’ conflict constraints to define the master relaxation and we check the feasibility of master solutions by using a *stabbing line algorithm* and a continuous non-convex optimization problem.

Using geographical and population data from IN-CORE (2022), we use the proposed model to determine optimal retrofitting and recovery actions in Joplin, MO. The results of the case study show that retrofitting actions should be performed in most locations in the geographical center of the city, which are also the locations with higher population density. They also show that spending budget in retrofitting is prioritized over recovery, and that only when a ‘critical’ set of locations in the center of the city are retrofitted, there should be expenditures in recovery. By performing simulations we also show that the proposed model outperforms other retrofitting policies, and that the optimal worst-case population dislocation is within 1% of the maximum population dislocation observed in the simulations.

To summarize, in this paper we make the following contributions:

- We propose a novel two-stage optimization model under uncertainty to aid decision-makers in the allocation of resources to retrofit and recover residential wood-frame buildings in tornado-prone regions.
- We explicitly model tornado paths as arbitrary line segments on the plane and formulate the problem as a two-stage robust optimization problem with integrality requirements in the first and second stages and in the uncertainty set.
- We develop an exact algorithm that embeds a decomposition branch-and-cut algorithm within a column-and-constraint generation method. By exploiting the geometric properties of line segments, we develop initialization and separation procedures to effectively implement the embedded decomposition branch-and-cut algorithm.

- Using real data we provide optimal retrofitting and recovery strategies for Joplin, MO. The results show that there can be up to 20% reductions in worst-case population dislocation by investing \$15 million and that our approach outperforms other retrofitting policies.

The remainder of this paper is organized as follows. Section 2 reviews the literature. Section 3 describes the two-stage robust optimization problem. Section 4 presents a customized version of the C&CG algorithm. We describe the decomposition branch-and-cut method to solve the subproblem in C&CG algorithm in Section 5. In Section 6, we conduct the numerical experiments on Joplin data as our case study. Lastly, Section 7 concludes the paper.

2. Literature review

2.1. Optimization models to retrofit and recover under tornado hazards

The determination of retrofitting and recovery strategies for tornado hazards, particularly using optimization, is a relatively new topic in the literature. Wen (2021) proposes a multi-objective optimization model to retrofit against tornado hazards. Their model, however, is single-stage, does not consider recovery, does not consider uncertain tornado paths, and aggregates all of the uncertainty of the problem into the parameters of the deterministic model. Simulation, on the other hand, is more commonly used to study the behavior of tornadoes, particularly to evaluate the impact of retrofitting strategies, see for example (Fan & Pang, 2019; Masoomi & van de Lindt, 2018; Stoner & Pang, 2021; Strader et al., 2016; Wang et al., 2017, 2021). These works, however, do not explicitly consider the decision problem of allocating resources to retrofit and recover.

The closest models in the literature to the present work are two-stage robust optimization models for (general) disaster planning, see Che et al. (2024), Cheng et al. (2021, 2024), Dang et al. (2023), Duran-Mateluna et al. (2023), Ma et al. (2018), Matthews et al. (2019), Velasquez et al. (2020), Wang and Zhong (2024), Yin et al. (2024), Yuan et al. (2016b). These models, however, are geared towards location, supply, and distribution problems in networks and do not capture the specific patterns of the disasters (such as, e.g., paths or ripple effects). Consequently, these models cannot be adapted to deal with tornadoes and the specific retrofitting and recovery setting we study here.

2.2. Robust optimization

Robust optimization methods have been extensively developed in the past two decades to mathematically model and study different systems under uncertainty. Standard robust optimization models assume that the uncertain parameters belong to a convex set (Ben-Tal & Nemirovski, 1999; Bertsimas & Sim, 2004), which have many applications such as scheduling (Lin et al., 2004), supply chain Ben-Tal et al. (2011), transportation (Yao et al., 2009), power system problems (Xiong et al., 2017), among others. Mixed-integer version of uncertainty sets have also been studied. Exact methods to solve a general robust optimization problem with a mixed-integer uncertainty set are similar to the integer L-shaped method (Laporte & Louveaux, 1993) in which a master problem iteratively solves a restricted version of the original problem under a few uncertain scenarios and for the solution vector. Here, a subproblem generates a new scenario by finding the worst-case of objective within a mixed-integer uncertainty set (Ansari et al., 2022; Ben-Tal et al., 2015; Borrero & Lozano, 2021; Ho-Nguyen & Kilinç-Karzan, 2018; Mutapcic & Boyd, 2009).

Ben-Tal et al. (2004) extend the scope of standard single-stage robust optimization problems by introducing the adjustable robust optimization methodology, where a part of decision variables must be determined before the realization of the uncertainty set. The rest

of the decision variables are chosen when the uncertain parameters are revealed. The two-stage approach provides a less conservative framework to deal with the uncertainty and it can model a broad range of applications in different areas such as transportation (Gabrel et al., 2014), networks (Atamtürk & Zhang, 2007), investment (Takeda et al., 2008), power systems problems (Li et al., 2021; Shams et al., 2021), among others.

In general, two-stage robust optimization problems are computationally expensive. To efficiently solve them, decomposition methods are widely employed, under the assumption that the second-stage problem is a linear programming problem (Gabrel et al., 2014; Jiang et al., 2012; Thiele et al., 2009; Zhao & Zeng, 2012). Zeng and Zhao (2013) presents the column and constraint generation method (C&CG) as an alternative to address this class of robust problems. They showed the C&CG can outperform previous decomposition for many problem settings. The C&CG approach has become a popular tool to solve two-stage robust optimization in the past decade (An & Zeng, 2015; An et al., 2014; Cheng et al., 2021; Dang et al., 2023; Ding et al., 2017; Duran-Mateluna et al., 2023; Jabr et al., 2015; Matthews et al., 2019; Neyshabouri & Berg, 2017; Velasquez et al., 2020; Yuan et al., 2016b), with several methodological enhancements being proposed recently (Dumouchelle et al., 2024; Goerigk & Kurtz, 2025; Subramanyam, 2022; Tsang et al., 2023; Wang et al., 2024). An important feature of existing applications of the C&CG method is that the second-stage optimization problem is convex. In contrast, in our problem the second-stage problem is an IP problem and the uncertainty set is mixed-integer non-linear. Therefore, the standard ‘dualize and combine’ approach that is used in the literature to solve the subproblem in the C&CG does not apply to our case.

3. Model formulation

Next, we define the two-stage robust optimization model to minimize the total population dislocation under an uncertain tornado and present a mathematical formulation of the problem.

3.1. Two-stage robust optimization formulation

Recall that the first stage decisions seek to determine what locations to retrofit before the realization of a tornado. The second stage decisions, that happen after the uncertainty is revealed, select what recovery strategies should be implemented in the locations that are affected by the tornado.

Formally, suppose S denotes the set of retrofitting strategies and P denotes the set of recovery plans. We consider a set of locations of interest L , hereafter referred to as locations for simplicity, which are placed in a 2-dimensional plane. Let $w_{\ell s}$ be the population dislocation estimation pre-tornado at location $\ell \in L$ under retrofitting strategy $s \in S$ (in most cases $w_{\ell s} = 0$) and let $g_{\ell sp}$ be the population dislocation post-tornado at location $\ell \in L$ assuming that the retrofitting strategy $s \in S$ is used and that the recovery plan is $p \in P$.

We assume that the decision-maker has a limited budget of $A \geq 0$ to invest in the retrofitting and recovery plans. This assumption allows the model to allocate the budget endogenously, rather than having the decision-maker setting the proportions before hand. If required, the model is able to handle fixed budgets by adding a linear constraint in the first-stage problem. Let $d_{\ell s}$ be the retrofitting cost of all the buildings in location $\ell \in L$ under strategy $s \in S$, and let $c_{\ell sp}$ denote the cost of using recovery plan $p \in P$ on the buildings in location $\ell \in L$ if the retrofitting strategy $s \in S$ is applied. All parameters in both stages are computed in Section 6.1 in a real example, by using the publicly available data and engineering models for wood-frame buildings in the literature.

Define the decision variables $f_{\ell s}$ and $r_{\ell sp}$ as

$$f_{\ell s} = \begin{cases} 1, & \text{if the buildings in location } \ell \in L \text{ are retrofitted using strategy } s \in S \\ 0, & \text{otherwise,} \end{cases}$$

$$r_{\ell sp} = \begin{cases} 1, & \text{if the buildings in } \ell \in L \text{ are retrofitted with strategy } s \in S \\ & \text{and recovered with plan } p \in P \\ 0, & \text{otherwise,} \end{cases}$$

and let z_ℓ , $\ell \in L$, be the binary variables that represent the coverage of a tornado across the locations, that is:

$$z_\ell = \begin{cases} 1 & \text{if location } \ell \in L \text{ is affected by the tornado} \\ 0 & \text{otherwise.} \end{cases}$$

Then, the two-stage robust optimization model is defined as follows

$$v = \min \sum_{\ell \in L} \sum_{s \in S} w_{\ell s} f_{\ell s} + \max_{z \in U} Q(z, f) \quad (1a)$$

$$\text{s.t. } \sum_{s \in S} f_{\ell s} = 1 \quad \forall \ell \in L \quad (1b)$$

$$f \in \{0, 1\}^{|L||S|}, \quad (1c)$$

where the second stage problem $Q(z, f)$ for given $z = (z_\ell : \ell \in L)$ and $f = (f_{\ell s} : \ell \in L, s \in S)$, is

$$Q(z, f) = \min \sum_{\ell \in L} z_\ell \sum_{s \in S} \sum_{p \in P} g_{\ell sp} r_{\ell sp} \quad (2a)$$

$$\text{s.t. } \sum_{\ell \in L} \sum_{s \in S} \sum_{p \in P} c_{\ell sp} r_{\ell sp} \leq A - \sum_{\ell \in L} \sum_{s \in S} d_{\ell s} f_{\ell s} \quad (2b)$$

$$\sum_{p \in P} r_{\ell sp} = f_{\ell s} \quad \forall \ell \in L, s \in S \quad (2c)$$

$$r \in \{0, 1\}^{|L||S||P|}. \quad (2d)$$

The first term of the objective function (1a) measures the pre-tornado dislocation across all locations due to implementing the retrofitting strategies. The second term measures the population dislocation after the worst-case tornado with respect to the retrofitting strategies selected at the first stage. Constraint (1b) ensures that each location picks exactly one retrofitting strategy. We make the assumption that there is a “do-nothing” strategy in S with zero cost.

The second stage optimization problem (2) selects the recovery vector f and the tornado coverage vector z . The budget constraint (2b) ensures that the overall cost for the execution of recovery and retrofitting strategies is A . Constraint (2c) forces each location to pick exactly one recovery strategy corresponding to the selected retrofitting strategy. As before, we assume there exist a “do-nothing” strategy in P with zero cost.

Note that, depending on the particular community, type of construction, and other characteristics, there may be different types of pre and post-event actions that decision-makers and stakeholders may take to mitigate or recover from a disruptive event such as a tornado. The proposed mathematical model was developed to be able to absorb this diversity of actions, through the use of sets of pre- and post-event decisions. This would enable, for example, to select among different roofing patterns, selection of construction materials, upgrades in construction codes, and others. In our case study in Section 6, without loss of generality, we select a reduced set of options that showcase the capability of our proposed model of selecting from a set of possible actions, while maintaining a compact size amenable to comparative analyses. Moreover, the proposed sets represent a realistic depiction of action that stakeholders may need to decide from, as funds for recovery may come from private or public sources. In this case, we aim to depict a scenario where decision-makers (e.g., local governments) want to determine the level of budget to allocate to provide immediate funding sources for recovery processes, as opposed to leaving all financial load to the individual building owner. To clarify, when referring to a do-nothing approach, the implication is not that the buildings will not be recovered, but that the decision-maker did not allocate funds for that particular building, potentially affecting the speed of its restoration.

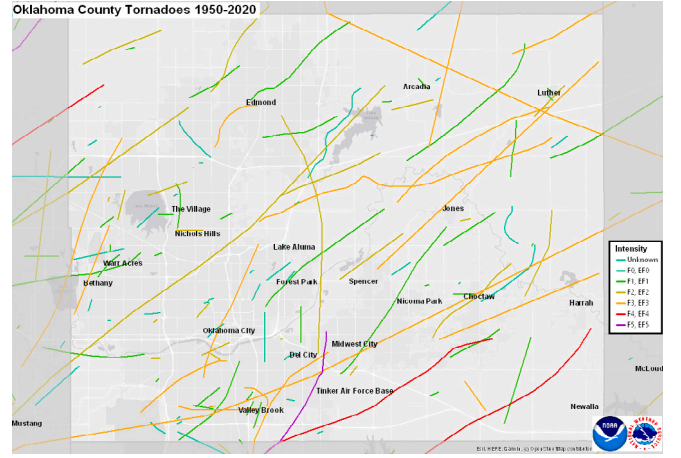


Fig. 1. Map of Oklahoma County tornado paths between 1950-2020 (NWC, 2023).

Next, we model tornado paths by ensuring that the decision vector z belongs to the uncertainty set $U \subseteq \{0, 1\}^{|L|}$ which contains all feasible tornado damages over locations in L .

3.2. Uncertainty set formulation

We make three assumptions to model a tornado and its associated damage:

- A.1. Tornado paths are represented by line segments of at most a length E in the (real) plane; $E > 0$ is a parameter set by the decision maker, and can be easily modified to study the effects of different tornado lengths.
- A.2. Each location $\ell \in L$ represents a cluster of buildings that are geographically close. Such clusters can depict individual buildings, blocks, neighborhoods, counties, or any other arrangement of interest to the decision-maker.
- A.3. A location $\ell \in L$ is damaged by the tornado if an aggregate measure of the coordinates of all buildings in ℓ (e.g., the centroid of all the coordinates of the buildings in that location), is within Δ units of the tornado path, $\Delta > 0$ being a parameter set by the decision-maker.

Assumption A.1. follows from considering historical records of tornadoes, such as evidenced in Fig. 1, which depicts the tornado paths in Oklahoma County between 1950 and 2020. Observe that, with a few exceptions, the paths can be adequately approximated by line segments. This behavior is also observed across the US, not only Oklahoma, see the damage assessment tool in NOAA (2023). Whereas A.1. might be seen as restrictive, it is a reasonable and uncluttered mathematical representation of a complex weather phenomenon.

Assumption A.2. is made to avoid politically inviable policies. Allowing full granularity (by associating a location with each building) might result in policies that prescribe retrofitting and recovery actions for a given building, but no retrofitting nor recovery to its immediate neighbors. Clearly, such a policy might raise fairness concerns, particularly if the decision-maker is a public entity. We note that our model works for any clustering approach; all clustering information required by the model is summarized via the parameters of the model.

Assumption A.3. defines the damage due to the tornado. Considering the variable nature of tornado damage, Δ can be seen as an upper-bound on the width observed in historical data for a given area to study the worst-case scenario. Due to the uncertain nature of tornado behavior, defining specific rules for modeling the coverage during a tornado remains a challenging task.

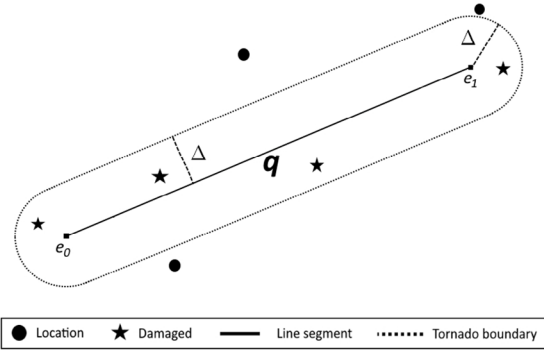


Fig. 2. The line segment q represents a tornado central line. Locations within Δ distance of the line segment (stars) are covered by the tornado.

We assume that the length of tornado q is at most a given parameter E , and that the two endpoints of q are located within a sufficiently large rectangle $R = [R_L^{(1)}, R_U^{(1)}] \times [R_L^{(2)}, R_U^{(2)}] \subseteq \mathbb{R}^2$ that contains all locations in L . Let $e_0 = (e_0^x, e_0^y) \in R$ and $e_1 = (e_1^x, e_1^y) \in R$ be the endpoints of line segment q (see Fig. 2). The uncertainty set $\mathcal{U}$ can be defined by the following mixed-integer non-linear programming (MINLP) formulation

$$\mathcal{U} = \left\{ z \in \{0, 1\}^{|L|} : \exists e_0, e_1 \in R, t \in \mathbb{R}^{|L|}, \right. \quad (3a)$$

$$\|e_0 + t_\ell(e_1 - e_0) - (x_\ell, y_\ell)\| \leq \Delta + M_\ell(1 - z_\ell), \quad \forall \ell \in L, \quad (3b)$$

$$0 \leq t_\ell \leq 1, \quad \forall \ell \in L, \quad (3c)$$

$$\|e_0 - e_1\| \leq E \Big\}, \quad (3d)$$

where $\|\cdot\|$ calculates the Euclidean distance between the vector coordinates and where (x_ℓ, y_ℓ) denote the coordinates of location $\ell \in L$ on the plane.

The value of $\|e_0 + t_\ell(e_1 - e_0) - (x_\ell, y_\ell)\|$ in Constraint (3b) calculates the distance between a point $e_0 + t_\ell(e_1 - e_0)$ on the line segment q and location ℓ . This constraint defines z -variables such that if $z_\ell = 1$ for some $\ell \in L$, then the location ℓ is covered by tornado. The value of M_ℓ is sufficiently large to ensure the constraint is trivially satisfied if $z_\ell = 0$ (the value of M_ℓ is set to be the largest value among the Euclidean distances from location $\ell \in L$ to the corners of rectangle R). Constraint (3c) enforces the range of continuous variable t_ℓ to be between 0 and 1, which indicates that q is a finite segment and not an infinite line. Constraint (3d) also ensures that the length of a tornado, which is the Euclidean distance between the two endpoints, does not exceed E . Note that the problem maximizes over the z -variables and that all of them have non-negative coefficients in the objective function. This observation and Constraint (3b) imply that $z_\ell = 1$ if and only if location $\ell \in L$ is covered by the tornado path.

Remark 1. The parameters Δ , E , $R_L^{(i)}$, and $R_U^{(i)}$, $i = 1, 2$, are selected by the decision-maker. Also, we note that $E = \infty$ is a valid choice for the length of the tornado path. In this case we refer to the path as a line rather than by a segment. This assumption represents situations where the path is sufficiently long to traverse the entire region of interest. In this case, constraints (3c) and (3d) are not necessary in the formulation of $\mathcal{U}$.

Remark 2. Additional assumptions on the tornado path reflecting other requirements of the decision-maker, can be included in the definition of the uncertainty set in (3) without increasing the modeling or computational complexity of the problem (see also Remark 5). Consider, for example, the following:

- Including specific locations: the decision-maker might be interested in the worst-case tornado that hits a set L^* of critical locations; for example, a set of hospitals, schools, power plants, or other critical infrastructure. This requirement can be included in (3) by adding constraints of the form $z_\ell = 1$ for all $\ell \in L^*$. Alternative requirements are also possible. For example, if the decision-maker is interested in the worst-case tornado that hits at least M of the locations in L^* one can alternatively include the constraint $\sum_{\ell \in L^*} z_\ell \geq M$ in (3).
- Including specific starting points: the decision-maker might want to consider the worst-case tornado that starts in a given circular region C . This requirement can be included by defining a new artificial location ℓ in the center of the circle and adding the constraint $z_\ell = 1$ in the definition of the uncertainty set. Observe that a tornado starts in C if and only if location ℓ is hit. An analogous procedure can be performed for fixed end points of the tornado path.

We close this section by noting that problem (1) belongs to the class of NP-hard problems, see Proposition 3 in Appendix 1. The proof follows by a reduction from the well-known *knapsack problem* and applies even if there is only one tornado path. Additionally, in Appendix 2, we have compiled a comprehensive nomenclature table summarizing the key terms and symbols.

4. A solution method based on the C&CG framework

We present a method to solve the two-stage robust optimization problem (1), based on the C&CG algorithm in Zeng and Zhao (2013). First, we provide a linear one-level reformulation of problem (1) by means of an epigraphic reformulation of the worst-case second-stage objective. To this end, note that $\mathcal{U}$ is a finite set and write it as $\mathcal{U} = \{z^1, \dots, z^n\}$, where $n \geq 1$ is the cardinality of $\mathcal{U}$ and suppose r^i is the vector of second-stage variables corresponding to scenario z^i . The one-level linear mixed-integer programming (MIP) reformulation of (1) is:

$$v = \min \sum_{\ell \in L} \sum_{s \in S} w_{\ell s} f_{\ell s} + \theta \quad (4a)$$

$$\text{s.t. } \sum_{s \in S} f_{\ell s} = 1 \quad \forall \ell \in L \quad (4b)$$

$$\theta \geq \sum_{\ell \in L} z_\ell^i \sum_{s \in S} \sum_{p \in P} g_{\ell s p} r_{\ell s p}^i \quad \forall i \in I \quad (4c)$$

$$\sum_{\ell \in L} \sum_{s \in S} \sum_{p \in P} c_{\ell s p} r_{\ell s p}^i + \sum_{\ell \in L} \sum_{s \in S} d_{\ell s} f_{\ell s} \leq A \quad \forall i \in I \quad (4d)$$

$$\sum_{p \in P} r_{\ell s p}^i = f_{\ell s} \quad \forall \ell \in L, s \in S, i \in I \quad (4e)$$

$$f \in \{0, 1\}^{|L||S|} \quad (4f)$$

$$r^i \in \{0, 1\}^{|L||S||P|} \quad \forall i \in I, \quad (4g)$$

where $I = [n] := \{1, \dots, n\}$. Constraint (4c) implies that the minimum value of θ is equal to the maximum population dislocation among all tornadoes in $\mathcal{U}$. Constraint (4d) and (4e) ensure that the recourse vectors r^i corresponding to each scenario, meet the constraints of the second stage.

Remark 3. Solving problem (4) directly is not practical in general, given that the uncertainty set $\mathcal{U}$ might have a large number of scenarios in terms of the number of locations. If $E = \infty$, then there is a polynomial number of scenarios in $\mathcal{U}$ which can be constructed based on the stabbing line algorithm in Section 5.3.1 (in this case, nevertheless, using (4) to directly solve the problem remains impractical because there would be at least a quadratic number of scenarios in terms of the number of locations). If $E < \infty$, it remains an open question to determine whether there are polynomially many scenarios in $\mathcal{U}$.

We employ a modified version of the C&CG to solve formulation (4). The algorithm is set to iteratively solve a relaxation of (4) over a subset $I^k \subseteq I$ of scenarios at iteration k to obtain an optimal solution (θ^k, f^k) . If given f^k , there exists a tornado coverage $z^{i_k} \in \mathcal{U}$ that produces a dislocation greater than θ^k , then we add scenario i_k to the master problem at the next iteration. Scenario i_k is found by solving the following subproblem, evaluated at f^k :

$$\Phi(f) = \max_{z \in \mathcal{U}} \min \left\{ \sum_{\ell \in L} z_{\ell} \sum_{s \in S} \sum_{p \in P} g_{\ell sp} r_{\ell sp} : r \in \mathcal{R}(f) \right\}, \quad (5)$$

where $\mathcal{R}(f)$ is the set of feasible solutions of problem $Q(z, f)$; see constraints (2b)–(2d). Specifically, if the optimal solution is such that $\Phi(f^k) > \theta^k$, then z^{i_k} is the tornado coverage z that attains $\Phi(f^k)$. Note that when the variables and constraints associated with scenario i_k are added at iteration $k+1$, the solution (θ^k, f^k) is no longer feasible in the master relaxation.

Let v^k be the value of the master problem at iteration k , that is, problem (4) with I replaced by I^k , with $I^k = \{i_0, i_1, \dots, i_{k-1}\}$ and let $\Phi(f^k)$ be the subproblem for the retrofitting strategy f^k found at iteration k of the algorithm. The C&CG method is presented in Algorithm 1.

Algorithm 1: C&CG algorithm to solve v

Data: Set $\mathcal{U}, L, S$, and P
Result: f^k
1 Set $k = 0$, $I^1 = \{i_0\} \subseteq \mathcal{U}$, $LB = -\infty$, and $UB = \infty$;
2 **while** $UB - LB > 0$ **do**
3 Set $k = k + 1$;
4 Solve the master MIP v^k and let f^k be the optimal retrofitting strategy. Update $LB = v^k$;
5 Solve the subproblem $\Phi(f^k)$ and let i_k be the index of the z -optimal solution in I . Update
 $UB = \min\{UB, \sum_{\ell \in L} \sum_{s \in S} \sum_{p \in P} \omega_{\ell s} f_{\ell s}^k + \Phi(f^k)\}$ and $I^{k+1} = I^k \cup \{i_k\}$;
6 **end**

Algorithm 1 begins with an arbitrary feasible scenario $i_0 \in I$. In each iteration k , the relaxed MIP problem v^k is solved. The value of v^k then provides a lower bound for the value of v . We note that the values of v^k , $k \geq 0$, are non-decreasing in k . Then, the subproblem $\Phi(f^k)$ generates a new feasible scenario i_k for the next iteration and updates the upper bound. The algorithm terminates when the subproblem does not find a scenario that violates the lower bound value, which implies the upper bound is equal to the lower bound. Zeng and Zhao (2013) prove that the C&CG algorithm converges in finite time.

Note that Step 5 in Algorithm 1 is expensive computationally because it requires solving the MINLP problem $\Phi(f)$ in (5). The approach to solve the subproblem in standard applications of the C&CG relies on using duality or the Karush Kuhn Tucker (KKT) conditions on the second-stage problem (Zeng & Zhao, 2013). Notice that $\Phi(f)$ is a bilevel problem with integrality requirements at both levels. Therefore, we cannot use either of these approaches. Alternatively, we propose a decomposition branch-and-cut method next.

5. A decomposition branch-and-cut (DBC) algorithm to solve $\Phi(f)$

The DBC method is based on a one-level reformulation of $\Phi(f)$. We replace the min in the objective by its epigraphic reformulation and replace the uncertainty set requirements in terms of linear (although exponentially many in the worst-case) constraints. Next, we discuss how to initialize the constraints in the master problem and how to perform the “two types” of separations.

5.1. Overview of the DBC algorithm

Let $z \notin \mathcal{U}$ be infeasible in the uncertainty set and define $C_z = \{\ell \in L : z_{\ell} = 1\}$. Observe that C_z is a set of locations that cannot be covered by a tornado path. Now, define $C = \{C_z : z \notin \mathcal{U}\}$ as the collection of

all such locations. A one-level linear reformulation of $\Phi(f)$ is given by:

$$\Phi(f) = \max \eta \quad (6a)$$

$$\text{s.t. } \eta \leq \sum_{\ell \in L} z_{\ell} \sum_{s \in S} \sum_{p \in P} g_{\ell sp} r_{\ell sp} \quad \forall r \in \mathcal{R}(f) \quad (6b)$$

$$\sum_{\ell \in C} z_{\ell} \leq |C| - 1 \quad \forall C \in \mathcal{C} \quad (6c)$$

$$z \in \{0, 1\}^{|L|}. \quad (6d)$$

The correctness of the reformulation follows from the maximization sense in (6). Note that r in (6b) represents a fixed value (not a variable); and that there is such an r for each element of $\mathcal{R}(f)$.

Observe that $\mathcal{R}(f)$ and $\mathcal{C}$ are finite sets with potentially exponentially many elements, thus $\Phi(f)$ cannot be solved directly by a commercial solver. The DBC starts with a relaxation of $\Phi(f)$ by replacing $\mathcal{R}(f)$ and $\mathcal{C}$ with subsets $\mathcal{R}^0(f)$ and $\mathcal{C}^0$, respectively. The resulting problem is referred to as the *master relaxation*. The standard branch-and-cut (BC) algorithm then starts solving the master and prunes nodes by bound and infeasibility. When at some node h an integral solution z^h is found (i.e., a solution that is feasible in the master), its feasibility with respect to the original formulation in (6) must be verified.

To this end, Algorithm 2 takes the master feasible solution (η^h, z^h) at node h of the BC tree and verifies its feasibility: first, it checks whether z^h satisfies all C -constraints (6c), i.e., it checks if $z^h \in \mathcal{U}$. If z^h does not pass the first check, then let $C_{z^h} = \{\ell \in L : z_{\ell}^h = 1\}$ be the *active locations* in z^h . Observe that C_{z^h} is a combination of locations that cannot be covered by a tornado path, thus a cut (6c) with $C = C_{z^h}$ is added on the fly. If z^h passes the first check, then the algorithm checks whether z^h satisfies all $\mathcal{R}(f)$ -constraints (6b). If z^h also passes the second check, then z^h is feasible in $\Phi(f)$; if not, then a cut (6b) with $r = r^h$ is added on the fly, r^h being the optimal solution of the second-stage problem associated with f and z^h . The performance of Algorithm 2 depends on the quality of $\mathcal{C}^0$ and $\mathcal{R}^0(f)$ and on having fast separation routines. We discuss these issues next.

Algorithm 2: Separation Procedure for the DBC

Data: Master feasible solution (η^h, z^h)

Result: Generating cuts on the fly, if (η^h, z^h) is infeasible

```

1 if  $z^h \notin \mathcal{U}$  then
2   Define  $C_{z^h} = \{\ell \in L : z_{\ell}^h = 1\}$ ;
3   Add constraint (6c) with  $C = C_{z^h}$  on the fly;
4 else if  $\eta^h > Q(z^h, f)$  then
5   Define  $r^h = \arg \min\{\sum_{\ell \in L} z_{\ell}^h \sum_{s \in S} \sum_{p \in P} g_{\ell sp} r_{\ell sp} : r \in \mathcal{R}(f)\}$ ;
6   Add constraint (6b) with  $r = r^h$  on the fly;
7 else
8    $(\eta^h, z^h)$  is feasible.
9 end

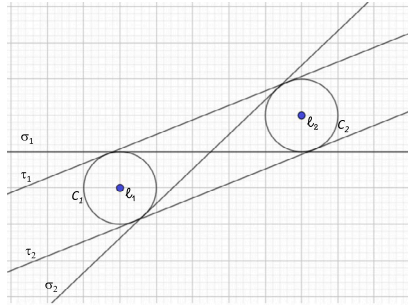
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5.2. Definition of $\mathcal{C}^0$

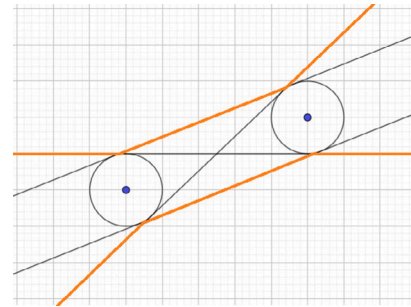
The set $\mathcal{C}^0$ is constructed by identifying pairs and triples of locations that cannot be covered by a single tornado path. It is worth mentioning that the valid constraints associated with these infeasible cases can also be used to tighten the relaxation of $\mathcal{U}$ by the elimination of potentially many infeasible fractional solutions for z -variables; we evaluate the performance of DBC with $\mathcal{U}$ enhanced by these valid cuts in Appendix 8.

5.2.1. Cuts from infeasible pairs

Because the length of a tornado is at most E and only locations within a distance of Δ from a tornado line segment are covered, a tornado cannot cover two locations $\ell_1, \ell_2 \in L$ if they are at a distance of more than $E + 2\Delta$. Consequently we have the following result:



(a) C_1 and C_2 are circles centered at ℓ_1 and ℓ_2 , respectively, with radius Δ ; τ_1 , τ_2 , σ_1 , and σ_2 are the (only) lines that are tangent to both C_1 and C_2 .



(b) The region $\tilde{P}_{\ell_1, \ell_2}$ is enclosed by the orange lines.

Fig. 3. Tangent lines to circles centered at ℓ_1 and ℓ_2 with radius Δ .

Proposition 1. Let $\Omega = \{(\ell_1, \ell_2) \in L^2 : \|(x_{\ell_1}, y_{\ell_1}) - (x_{\ell_2}, y_{\ell_2})\| > 2\Delta + E\}$ be the set of infeasible pairs. Then any $z \in \mathcal{U}$ satisfies that

$$z_{\ell_1} + z_{\ell_2} \leq 1 \quad \forall (\ell_1, \ell_2) \in \Omega. \quad (7)$$

The proof of Proposition 1 is given in Appendix 3.

5.2.2. Infeasible triple cuts

We also identify infeasible triples of locations that cannot be covered by a tornado path. To introduce the infeasible set of triples, we assume $E = \infty$ which implies a line representation for a tornado path. Observe that if a line does not cover a subset of locations $Z \subseteq L$, then there is no segment of the line that covers the locations in Z either. For any locations ℓ_1 and ℓ_2 , the analysis below partitions the plane ($\mathbb{R}^2$) in two regions, denoted by R'_{ℓ_1, ℓ_2} and R''_{ℓ_1, ℓ_2} . Region R'_{ℓ_1, ℓ_2} is the set of points $\ell \in \mathbb{R}^2$ for which there exist a line that covers ℓ_1, ℓ_2 and ℓ , and $R''_{\ell_1, \ell_2} = \mathbb{R}^2 \setminus R'_{\ell_1, \ell_2}$. For simplicity, in the next discussion, we interchangeably refer to a point in $\mathbb{R}^2$ by its name, as $\ell \in \mathbb{R}^2$, or by its two-dimensional coordinates, as in $(x_\ell, y_\ell) \in \mathbb{R}^2$.

Observe that a line $\lambda \subset \mathbb{R}^2$ covers both ℓ_1 and ℓ_2 if and only if $\lambda \cap B_1 \neq \emptyset$ and $\lambda \cap B_2 \neq \emptyset$, where B_i is the ball with center ℓ_i and radius Δ , $i = 1, 2$. Let $\mathcal{L}_{\ell_1, \ell_2}$ be the collection of these lines, and let P_{ℓ_1, ℓ_2} be the set of points that belong to any such line:

$$P_{\ell_1, \ell_2} = \{\ell \in \mathbb{R}^2 : \exists \lambda \in \mathcal{L}_{\ell_1, \ell_2} \text{ s.t. } \ell \in \lambda\}. \quad (8)$$

For any set $T \subseteq \mathbb{R}^2$ let $T(\Delta)$ be the set of points that are at a distance at most Δ from T , that is

$$T(\Delta) = \{(x_\ell, y_\ell) \in \mathbb{R}^2 : \exists (x_t, y_t) \in T \text{ s.t. } \|(x_\ell, y_\ell) - (x_t, y_t)\| \leq \Delta\}. \quad (9)$$

It is readily seen that $R'_{\ell_1, \ell_2} = P_{\ell_1, \ell_2}(\Delta)$.

Now, the set P_{ℓ_1, ℓ_2} can be characterized by the four lines τ_1 , τ_2 , σ_1 and σ_2 that are tangent to both C_1 and C_2 , where C_1 and C_2 are the circles enclosing B_1 and B_2 , see Fig. 3(a). Indeed, let $\tilde{P}_{\ell_1, \ell_2}$ be the set enclosed by these tangents, see Fig. 3(b), in Appendix 4 we show that $\tilde{P}_{\ell_1, \ell_2} = P_{\ell_1, \ell_2}$.

Given the previous considerations, in order to characterize R'_{ℓ_1, ℓ_2} it is necessary to characterize $\tilde{P}_{\ell_1, \ell_2}(\Delta)$. From the shape of $\tilde{P}_{\ell_1, \ell_2}$, it is clear that $\tilde{P}_{\ell_1, \ell_2}(\Delta)$ looks very similar to $\tilde{P}_{\ell_1, \ell_2}$, with the difference that its border starts Δ units up and Δ units down, see Fig. 4(a). The equations of the lines in orange defining $\tilde{P}_{\ell_1, \ell_2}(\Delta)$, i.e., the equations of the lines $\lambda_1, \dots, \lambda_6$ in Fig. 4(b), are derived in Appendix 4.

Now, let

$$\Gamma_{\ell_1, \ell_2} = \{\ell \in L \setminus \{\ell_1, \ell_2\} : (x_\ell, y_\ell) \in \mathbb{R}^2 \setminus \tilde{P}_{\ell_1, \ell_2}(\Delta)\}. \quad (10)$$

The previous discussion is summarized in the following result, which states that if a location belongs to the infeasible set Γ_{ℓ_1, ℓ_2} then at most

two locations among ℓ_1, ℓ_2 , and ℓ_3 can be chosen to be covered by a tornado.

Proposition 2. Let ℓ_1 and ℓ_2 be two elements of L and suppose that $D(\ell_1, \ell_2) > 2\Delta$. Then any $z \in \mathcal{U}$ satisfies that

$$z_{\ell_1} + z_{\ell_2} + z_\ell \leq 2 \quad \forall \ell \in \Gamma_{\ell_1, \ell_2}. \quad (11)$$

It is worth mentioning that we do not need to consider all possible combinations of locations to construct the infeasible triple cuts because if $\ell_3 \in \Gamma_{\ell_1, \ell_2}$, then $\ell_1 \in \Gamma_{\ell_2, \ell_3}$ and $\ell_2 \in \Gamma_{\ell_1, \ell_3}$. Also, we note that the pair-wise and triple valid constraints introduced so far do not completely characterize $\mathcal{U}$, as shown in the next remark. However, they can be used to define the initial subset $\mathcal{C}^0 \subseteq \mathcal{C}$ of the master relaxation in the DBC as follows:

$$\mathcal{C}^0 = \{(\ell_1, \ell_2) \in \Omega\} \cup \{(\ell_1, \ell_2, \ell_3) : \ell_3 \in \Gamma_{\ell_1, \ell_2}, \ell_1, \ell_2 \in L, D(\ell_1, \ell_2) > 2\Delta\}. \quad (12)$$

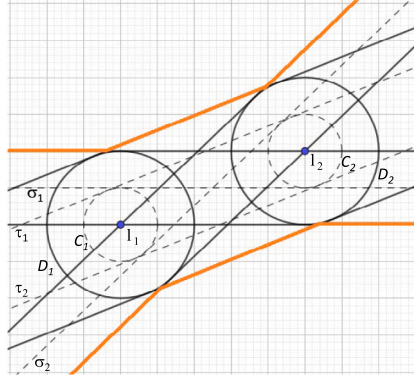
Remark 4. Define $\mathcal{V} = \{z \in \{0, 1\}^{|L|} : \sum_{\ell \in C} z_\ell \leq |C| - 1, \forall C \in \mathcal{C}^0\}$. A solution determined by the set $\mathcal{V}$ is not necessarily feasible to $\mathcal{U}$; specifically $\mathcal{U} \subset \mathcal{V}$ and in general there exist points in $\mathcal{V}$ that do not belong to $\mathcal{U}$. For example, consider a case of three locations ℓ_1, ℓ_2 , and ℓ_3 with coordinates $(0, 0)$, $(4, 0)$, and $(2, 1.1)$ on a plane under a tornado with the maximum length of $E = 2$ and the width of $\Delta = 1$ (See Fig. 5). It can be easily observed that there exists no tornado path that covers all three locations because the only tornado path covering ℓ_1 and ℓ_2 is the line segment with endpoints $(1, 0)$ and $(3, 0)$. Obviously, the location $(2, 1.1)$ is not covered by this path. However, $z_{\ell_1} = z_{\ell_2} = z_{\ell_3} = 1$ is a feasible solution in $\mathcal{V}$ because it satisfies both the infeasible pair and triple cuts.

5.3. Feasibility check in $\mathcal{C}$

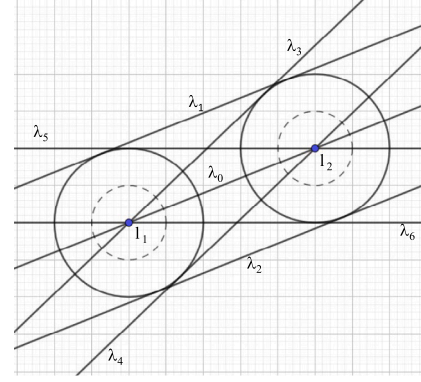
Next, we present a method to determine whether a master feasible solution z^h belongs to $\mathcal{U}$. We start our analysis for the case when $E = \infty$ and then present a method for the case when $E < \infty$.

5.3.1. Feasibility check for $E = \infty$ using the stabbing line algorithm (SLA)

The stabbing line problem consists of a collection of (disjoint) circles $\mathcal{O} = \{O_1, O_2, \dots, O_n\}$ and the objective is to find the line on the plane that intersects the most circles. The SLA is a polynomial time algorithm that solves this problem and reports $c(\mathcal{O})$: the maximum number of circles in $\mathcal{O}$ that can be intersected with a line. Consider the collection of circles $\mathcal{O}^h$ with centers at the locations in C_{z^h} and radius Δ . Observe that a tornado path with $E = \infty$ covers all locations in C_{z^h} if and only



(a) The region $\tilde{P}_{\ell_1, \ell_2}(\Delta)$ is enclosed by the orange lines. The tangents τ_1 , τ_2 , σ_1 , and σ_2 , as well as the circles C_1 and C_2 are shown in dashed lines. The circles D_1 and D_2 centered at ℓ_1 and ℓ_2 , respectively, have radius 2Δ .



(b) The lines λ_i , $i = 0, \dots, 6$, that define the region $\tilde{P}_{\ell_1, \ell_2}(\Delta)$. The smaller circles are C_1 and C_2 , the larger circles are D_1 and D_2 .

Fig. 4. The region $\tilde{P}_{\ell_1, \ell_2}(\Delta)$.

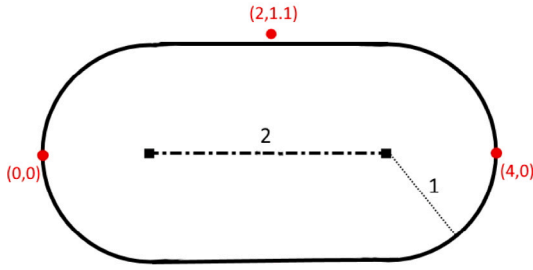


Fig. 5. The only possible tornado path covering buildings (0,0) and (4,0).

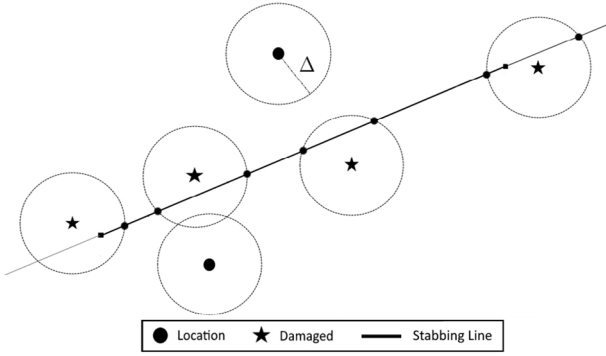


Fig. 6. Finding the locations that are covered by a tornado within distance Δ is equivalent to finding a line that stabs the circles centered at the locations with radius Δ .

if $c(\mathcal{O}^h) = |C_{z^h}|$ (see Fig. 6). Therefore, if $c(\mathcal{O}^h) < |C_{z^h}|$, then $z^h \notin \mathcal{U}$ whereas if $c(\mathcal{O}^h) = |C_{z^h}|$ then $z^h \in \mathcal{U}$. The details behind the stabbing line algorithm are in Appendix 6.

5.3.2. Feasibility check if $E < \infty$

In this case, if $c(\mathcal{O}^h) < |C_{z^h}|$, then we can again conclude that there is no finite line segment that covers C_{z^h} and thus $z^h \notin \mathcal{U}$. Otherwise, if $c(\mathcal{O}^h) = |C_{z^h}|$, then there is an infinite line that covers all locations in C_{z^h} , but there is no guarantee that a line segment of length at most E covers all locations of C_{z^h} . Thus, we complete the feasibility check in two steps: first, we use the line resulting from the SLA and projection

ideas to determine whether the line can be trimmed to be of length E without compromising its coverage, the details are in Appendix 5. If even after this step the feasibility of z^h cannot be guaranteed, then we check the feasibility of z^h algebraically using a simplified version of $\mathcal{U}$.

Specifically, consider the set $\tilde{\mathcal{U}}^h$ defined as

$$\tilde{\mathcal{U}}^h = \left\{ (e_0, e_1) \in \mathbb{R}^2 : \exists t \in [0, 1]^{|C_{z^h}|}, \|e_0 - e_1\| \leq E \right. \quad (13a)$$

$$\left. \|e_0 + t_\ell(e_1 - e_0) - (x_\ell, y_\ell)\| \leq \Delta, \forall \ell \in C_{z^h} \right\}. \quad (13b)$$

Observe that $z^h \in \mathcal{U}$ if and only if $\tilde{\mathcal{U}}^h \neq \emptyset$ and that (13) is far simpler than $\mathcal{U}$ because it has fewer variables and does not involve binary variables nor ‘big-M’ values. Checking whether $\tilde{\mathcal{U}}^h$ is non-empty can be done using a quadratic (non-convex) continuous optimization solver. The such check can be significantly speed-up by adding linear cuts in $\tilde{\mathcal{U}}^h$, see the details in Appendix 5.

5.4. Definition of $\mathcal{R}^0(f)$ and feasibility check for $\mathcal{R}(f)$

To initialize the master problem we assume that $\mathcal{R}^0(f) = \{r^0\}$ where r^0 is an optimal solution for the second-stage problem $Q(z, f)$ assuming no location is hit, i.e., r^0 is a solution of $Q(0, f)$. On the other hand, once it has been checked that $z^h \in \mathcal{U}$, we check that z^h satisfies all $\mathcal{R}(f)$ -constraints in (6b) by solving the second-stage problem evaluated at z^h , i.e., by solving $Q(z^h, f)$. Let r^h be an optimal solution of $Q(z^h, f)$. If $\eta^h > Q(z^h, f)$ then (η^h, z^h) is not feasible in (6b) and we add the constraint $\eta \leq \sum_{\ell \in L} z_\ell \sum_{s \in S} \sum_{p \in P} g_{\ell s p} r_{\ell s p}^h$ at all active nodes of the BC tree. Conversely, if $\eta^h \leq Q(z^h, f)$, then (η^h, z^h) is a feasible solution of (6) and node h is pruned by feasibility after updating the lower bound of the BC. Observe that the initialization and separation procedure involve solving the NP -hard problem $Q(z, f)$; however, this problem is relatively simple and can be solved quickly using commercial MIP solvers, see our computational results in Section 6.

Remark 5. Consider the extensions of the uncertainty set described in Remark 2. Computationally, in either case the only modification that would need to be done in the C&CG algorithm would be to include the additional constraints in problem (6) and in the corresponding master relaxation problem.

Table 1
Retrofitting strategies for the experiments.

Strategy s	Description
R_1	Roof covering with asphalt shingles, using 8d nails to attach roof deck sheathing panel to rafters spaced at 6/12 inch, and the selection of two 16d toenails for roof-to-wall
R_2	Roof covering with asphalt shingles, using 8d nails to attach roof deck sheathing panel to rafters spaced at 6/6 inch, and the selection of two H2.5 clips for roof-to-wall
R_3	Roof covering with clay tiles, using 8d nails to attach roof deck sheathing panel to rafters spaced at 6/6 inch, and the selection of two H2.5 clips for roof-to-wall.

6. Case study: retrofitting/recovery residential buildings in Joplin, MO

Next, we perform several numerical experiments with the proposed model. Our objectives are to study the performance of the two-stage optimization problem, to analyze how the optimal solutions change with the length of the path, to gather high-level insights into the optimal retrofitting and recovery activities, and to compare the optimal policies with other benchmark policies. We also show that using infeasible triple cuts is key to find optimal solutions in a timely manner. Finally, we consider an alternative model when recovery can be fractional instead of just discrete. The experiments are based on location and cost data from Joplin, MO. Joplin was the location of one of the most severe tornadoes in US history that occurred on May 22, 2011. The tornado was rated EF-5 with an average speed of more than 200 mph and crossed 22.1 miles on the ground with a width of one mile. This catastrophe caused the loss of 161 lives and more than a thousand injuries. The total damage was approximated at 3 billion dollars as around 7500 residential structures were destroyed (Kuligowski et al., 2014).

6.1. Definition of parameters

The geographic and demographic data are based on information from Joplin and extracted from the online platform [IN-CORE \(2022\)](#). Particularly, this dataset divides buildings in about 1500 “block IDs” (which are contiguous blocks of the city) and includes over 20,000 buildings. For our experiments, we use only the single or multi-story residential wood-frame building archetypes.

We assume an EF-2 tornado with a wind speed of 134 mph, a width of 0.75 miles (thus $\Delta=0.75/2$), and a length of 5 and ∞ miles. Following Wang et al. (2021), we consider the retrofitting strategies presented in Table 1; these strategies are analyzed in Masoomi et al. (2018) with respect to the component fragility of wood-frame residential building archetypes. The retrofitting activities in Table 1 have minimal impact in the livability of the buildings; thus we assume that the first stage population dislocation parameter is zero for all locations and strategies, that is, $w_{\ell s} = 0$ for all $\ell \in L$, $s \in S$. This assumption, however, does not change the complexity of solving the problem, and if necessary could be relaxed to better depict a particular context or instance of interest. The cost of implementing these strategies are computed from the data of [IN-CORE \(2022\)](#). As mentioned in Section 3, a ‘do-nothing’ strategy with zero cost is also included in the retrofitting strategy set S , that is, $S=\{\text{do-nothing}, R_1, R_2, R_3\}$.

We consider only two recovery strategies, namely, recover ($p = 1$) and do-nothing ($p = 0$); thus $P = \{0, 1\}$ and total budgets of 0, 15, and 30 million dollars. For our experiments we define $g_{\ell sp}$ as the expected population dislocation 60 days after the tornado hits location ℓ if retrofitting strategy s and recovery plan p are implemented. The calculation of $g_{\ell sp}$ depends on the level of damage produced by the tornado, as well as the effects of retrofitting and recovery in each specific location. The detailed computation of these parameters using

real and engineering data, as well as the computation of the costs of retrofitting and repair, are described in detail in Appendix 7.

6.2. Robust retrofitting and recovery strategies in Joplin, MO

We implement the proposed two-stage optimization model to find robust retrofitting and recovery strategies for locations in Joplin. All algorithms are coded in Python 3.9 and solved using Gurobi Optimizer 9.1.1 on a 64-bit Windows operating system with Intel(R) Core(TM) i7-8550U CPU and 8.00 GB of RAM under a time limit of 3600 s. We use Algorithm 1, along with the DBC method described in Section 5 with the setup that yields the best performance; see our comparative experiments in Table 9 in Appendix 7.

In order to define the locations in L , we group Joplin “block IDs” in 100, 150, and 200 center locations using the k -means method (MacQueen, 1967) and aggregate the data accordingly. Particularly, we assume that when the tornado hits a location then all of its buildings are hit, and that the tornado hits a location only if the location’s centroid is within Δ units of the path. In this experiment, we reduce the size of the initial optimization model in the subproblem by generating *infeasible triple cuts* only for constraints that involve the top 20% of locations with the highest population. Table 2 compares the population dislocation when the tornado length is $E = 5$ miles or ∞ and the budgets are $A = 0, 15, 30$ million USD. Column 1 in Table 2 has the number of center locations that is set for clustering by k -means method. Columns 2 and 3 show the values for the parameters E and A , respectively. Column 4 has the optimal population dislocation under the worst-case tornado for each set of parameters. Columns 5 and 6 report the CPU run time of C&CG Algorithm 1 in seconds and its number of iterations, respectively. Column 7 shows the CPU runtime of the DBC method in seconds to solve subproblem $\Phi(f)$. Column 8 compares the CPU time that the DBC algorithm spends in a callback function to verify the feasibility of integral solutions. Column 9 shows the total CPU time to find a feasible solution in the set $\tilde{U}^h$. Columns 10 and 11 present the proportion of the total budget that is spent for retrofitting and recovery, respectively.

Table 2 shows that the solver is able to find optimal solutions for the 100 and 150 center locations experiments in at most 276 and 2274 s, respectively. It also shows that solving the subproblems is the most time-consuming step (which is virtually 99% of the total runtime).

As expected, the C&CG algorithm runtime exponentially increases with the number of locations. For the experiment with 200 locations, the optimal solutions were not achieved for some parameters within the time limit. In these cases, the percentage value in brackets indicates the optimality gap at the time of termination; it is important to note that the algorithm is allowed to complete its current iteration even after surpassing the time limit.

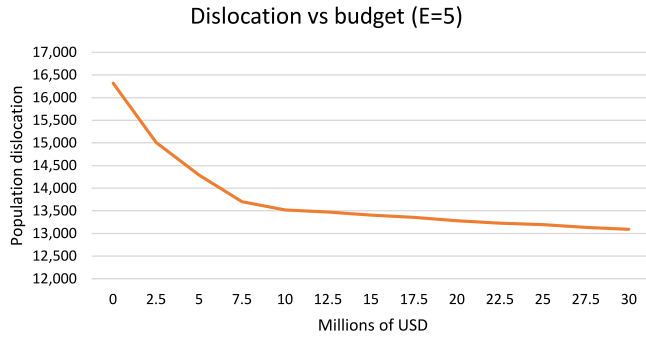
For all experiments, the callback function (i.e., Algorithm 2) is very fast, which implies that the bottleneck of the entire algorithm is the branch-and-bound that solves the relaxation of the subproblem. As expected, for the infinite line, the results show zero seconds to check the feasibility using $\tilde{U}^h$, because in this case, the SLA alone can verify the feasibility.

For both tornado lengths, there are decreases in the dislocation by increasing the budget because decision-makers are able to retrofit and recover more locations. For example, for 100 locations, increasing the budget from \$0 to \$15M results in around 3000 fewer dislocations in damaged neighborhoods. Note that this translates in reductions of around 20% in population dislocation for both tornado lengths by investing \$15M. However, adding another \$15 million will reduce dislocations only by around 500 people because of the great cost of recovery. In fact, the relationship between budget and dislocation exhibits a clear diminishing returns behavior, see Fig. 7, where it can be seen that after around \$10M there is a small decrease in population dislocation for 100 locations. From a decision-making point of view,

Table 2

Solving the two-stage robust optimization model with different parameters of locations, tornado length, and available budget, after 60 days of recovery in Joplin, MO.

# center locations	Length	Budget (\$)	Population dislocation	CCG run time (s.)	CCG iteration	DBC run time (s.)	Callback run time (s.)	$\tilde{U}^h$ feasibility run time (s.)	Budget proportion for retrofitting	Budget proportion for recovery
100	5	0 M	16 318	94	2	94	10	2	–	–
		15 M	13 408	131	2	131	9	3	0.6	0.4
		30 M	13 091	276	3	275	22	9	0.3	0.7
	∞	0 M	17 293	71	2	70	6	0	–	–
		15 M	14 236	158	3	157	9	0	0.6	0.4
		30 M	13 839	146	3	145	11	0	0.4	0.6
150	5	0 M	14 850	674	2	674	8	1	–	–
		15 M	12 256	2272	5	2270	18	5	0.8	0.2
		30 M	11 945	2274	4	2273	22	7	0.4	0.6
	∞	0 M	15 989	595	2	595	15	0	–	–
		15 M	13 142	1326	4	1325	16	0	0.8	0.2
		30 M	12 757	1208	4	1208	20	0	0.4	0.6
200	5	0 M	15 002	2559	2	2558	17	3	–	–
		15 M	12 754	4750(5%)	2	4750	22	6	0.6	0.4
		30 M	12 539	4736(6%)	2	4736	29	11	0.3	0.7
	∞	0 M	15 974	1630	2	1630	9	0	–	–
		15 M	13 135	3947(2%)	3	3947	13	0	0.7	0.3
		30 M	12 937	4039(2%)	3	4038	17	0	0.4	0.6

**Fig. 7.** Budget vs population dislocation for a path length of $E = 5$ miles through 100 locations.

these results imply that it is not reasonable to spend more than \$10M–\$15M in retrofitting and recovery, as the return of investment is not worth it.

In the experiments with 150 and 200 locations, the population dislocation values are generally lower compared to the 100 location scenario. This can be attributed to fewer buildings per location with less retrofitting and recovery costs, which allows for more targeted budget allocation. For example, we observe that for the case of with $E = 5$ miles and \$30M budget, the optimal solution of 100 locations left \$737K from the total budget, whereas the remaining budget for 150 and 200 locations are \$217K and \$31K out of \$30M, respectively. Based on this observation, we suspect that more granular clustering results in better budget allocations and less population dislocation. We note however, that the level of clustering or granularity depends on the decision-maker, and that having a fully granular model might not be desirable due to practicality or fairness considerations.

More details on the optimal strategies can be found in Fig. 8, which maps the optimal results for the case with 100 locations and the worst-case tornado scenarios for various budgets. It shows that in all experiments the decision-maker uses most of the extra amount of budget from \$15M to \$30M to recover a few locations rather than in retrofit locations that are not in the path of the worst-case tornado. Notice that recovery costs are generally more expensive than retrofitting costs. For example, the recovered location in the ∞ –\$15M map in Fig. 8 has 470 residents; 382 of which are estimated to be dislocated by spending \$194K for retrofitting. However, the decision-maker has to spend \$5.5M in recovery to further reduce the estimated dislocation

to 295 people for this location. Columns 10–11 in Table 2 show that the proportion of the budget that is assigned to recovery increases by having more budget. In addition, we can observe in Fig. 8 that the path of worst-case tornadoes does not change by much in different experiments, suggesting that the location of the tornado path is largely driven by the population density in the city center area.

Note that optimal solutions might include retrofitting particular areas that ultimately escape the impact of the worst-case tornado (blue squares in Fig. 8). Failure to allocate resources for retrofitting locations identified beyond the reach of the worst-case tornado in an optimal solution may lead to a suboptimal outcome where the path of the worst-case tornado shifts to encompass non-retrofitted areas, resulting in a more extensive population dislocation. It is also worth noting that, in general, recovery actions tend to result in much larger costs compared to retrofitting actions. Unfortunately, the literature shows that despite it being a more cost-effective investment, not enough mitigation actions are taken to reduce the social and economic costs related to disruptive events. Nevertheless, it is important to mention that in our particular case study, we are evaluating the effects of a single disruptive event, but in general, once a building has been retrofitted, it will significantly reduce its damage and recovery costs for multiple disruptive events (for example, in locations where wind damages occur frequently).

These results suggest, from a worst-case perspective, that it is optimal to invest only in locations in or close to where the worst-case tornado hits and that it is better to invest both in retrofitting and recovery, rather than just in retrofitting. The results are somewhat surprising in the sense that, at least a priori, it would be reasonable to invest all budget in retrofitting across a wider area (particularly because recovery is much more expensive than retrofitting). As such, the results suggest that there is a critical set of clustered locations, in this case, the densely populated center of the city, where most of the investments must be concentrated. We suspect, however, that there is a different structure for the optimal actions in data sets with a more homogeneous population distribution.

6.3. Advantages of robust retrofitting

In this section, we compare the optimal two-stage robust retrofitting and recovery strategies against two policies: a *random* policy that makes retrofitting decisions at random; and a *greedy* policy that greedily allocates budget using as criteria the frequency at which locations are hit and the total population of each location. For these experiments, we assume $E = 5$ miles, 100 locations, and a total budget of $A = \$15$ million.

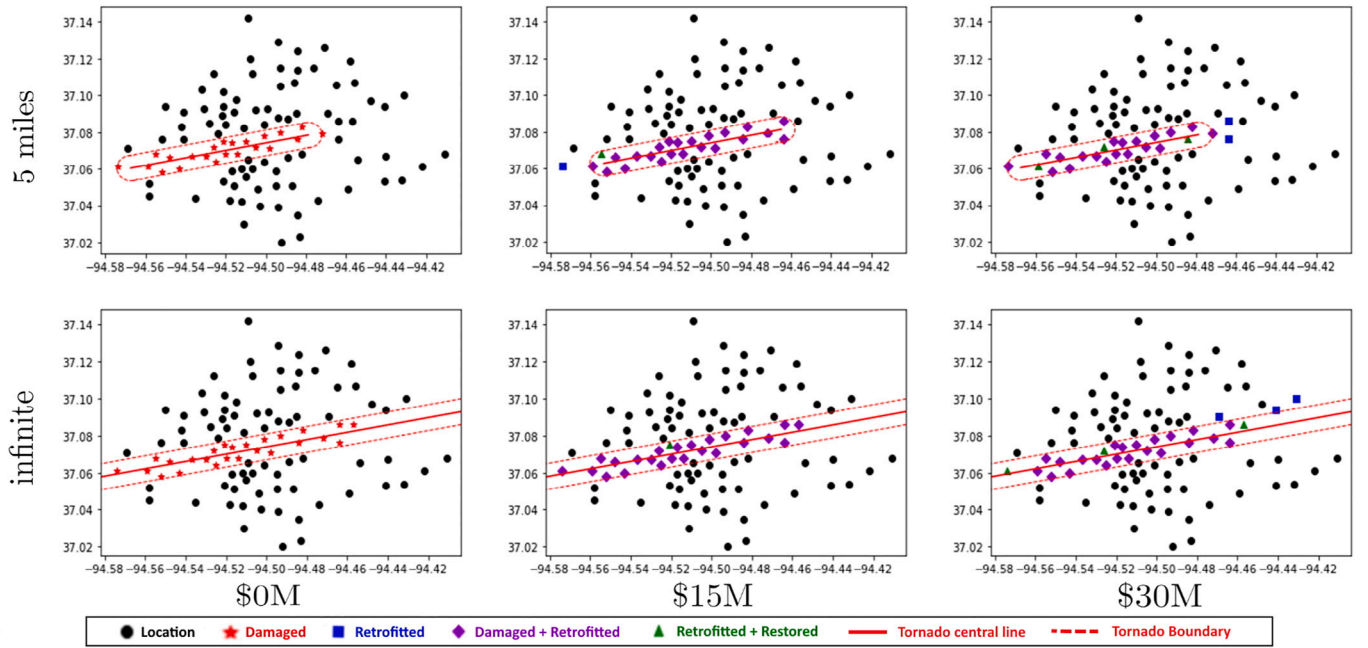


Fig. 8. Maps of 100 locations and their retrofitting and recovery strategies under the worst-case tornado scenario for six parameter settings in Table 2. When retrofitting, the selected strategy is almost exclusively R_3 , which is the best, but most expensive, strategy.

Table 3

Comparison of worst-case population dislocation statistics for random retrofitting and greedy retrofitting plans by spending a certain amount of budget in retrofitting out of a total \$15M. The statistics for the random retrofitting are over 50 randomly generated solutions, for each given budget.

Budget to retrofit	Random Retrofitting				Greedy Retrofitting
	Average	Maximum	Minimum	Standard deviation	
\$0M (0%)	16 318			–	16 318
\$3M (20%)	16 067	16 318	15 703	188	14 956
\$9M (60%)	15 606	16 084	14 949	265	14 069
\$15M (100%)	15 231	15 962	14 529	331	13 522
Robust optimal retrofitting	13 408				

For the random policy, we assume that the decision-maker is allowed to spend \$0, \$3, \$9, and \$15 million of the budget in each experiment to randomly retrofit locations with random selections in set S ; the remaining budget is reserved for the recovery actions. Once a retrofitting strategy f is determined, then the corresponding worst-case second stage happens (i.e., the value of $\Phi(f)$ is determined, see Eq. (5)).

For the greedy policy, we first simulate 50 tornado scenarios over the region of interest. We then identify the two scenarios among the 50 simulated tornadoes that result in the most population dislocation (i.e., the two worst scenarios) and collect the locations that are hit by these tornadoes. In the next step, we rank the selected locations based on the number of hits by all simulated tornadoes. Locations with a higher frequency of hits will be prioritized to receive retrofitting as they are identified as more vulnerable locations to tornadoes by simulation results. Similar to the random retrofitting experiment, we assume that the decision-maker is allowed to spend \$0, \$3, \$9, and \$15 million out of \$15 million budget to retrofit the selected locations. As with the random policies, once a retrofitting strategy f is determined, then the corresponding worst-case second-stage $\Phi(f)$ is computed.

Table 3 compares the policies. For the random policy, given a budget, we generate 50 retrofitting solutions at random. The average, maximum, minimum, and standard deviation of the worst-case population dislocation across these solutions are shown in columns 2–5 of Table 3 (note that for the \$0M case, we do not need multiple replications.). The last column of the table has the worst-case population dislocation corresponding to the greedy policy.

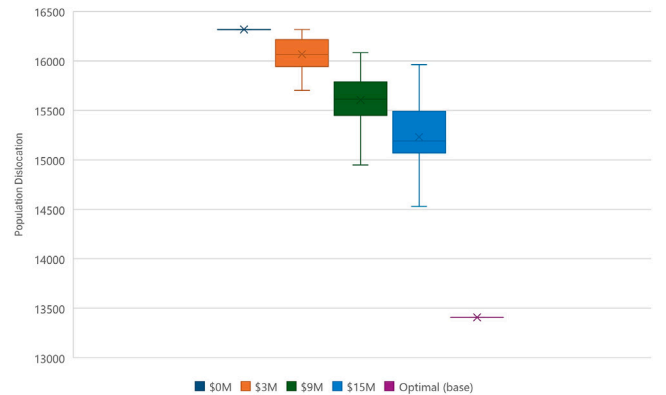


Fig. 9. Box and Whisker plots for random policy in Table 3 experiments. It shows the reduction trend for worst-case population dislocation by considering more budget for retrofitting.

The results in Table 3 show that with random policies the average, maximum, and minimum dislocation after the corresponding worst-case tornado decrease with the percentage of the budget spent in retrofitting. These results are depicted graphically in the Box plots in Fig. 9. Note that the proposed robust optimization model results in 13408 dislocations, which is noticeably better than the random policies.

Table 4

Comparison of population dislocation statistics using ten different random retrofitting plans and budgets of \$15M and \$30M.

Budget	Population dislocation					
	Optimal	Greedy	Average	Maximum	Minimum	Standard deviation
\$15M	13 408	13 522	15 314	15 941	14 545	331
\$30M	13 091	13 100	14 795	15 248	14 145	294

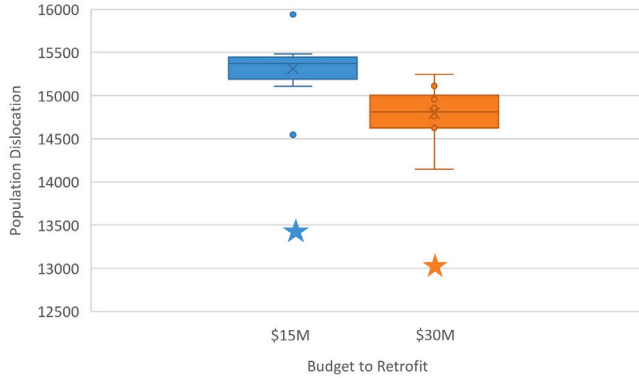


Fig. 10. Box and Whisker plots for Table 4 experiments. It shows the gap between the robust optimal plan (stars) and the population dislocation of the random retrofitting strategies for the base worst-case tornado.

To further emphasize the value of optimization, recall that the optimal solution with our proposed method in Table 2 spends \$9M out of \$15M (60% of budget) in retrofitting. By contrast, even the minimum population dislocation with random retrofitting using the same budget proportion is 14949 (a 11% increase).

Table 3 also shows that the greedy retrofitting policy leads to less worst-case population dislocation compared to all implemented random retrofitting policies across all budgets. The robust optimal retrofitting with a value of 13408 is still the best policy to minimize worst-case population dislocation; however, the performance of the greedy policy is remarkably close to that of the optimal policy, particularly when all budget is spent on retrofitting. This result suggests that using the greedy policy that spends all the budget in retrofitting, might be a reasonable proxy for the worst-case optimal policy.

We conduct additional simulations to randomly retrofit locations with the full amount of \$15M and \$30M budget as before, but this time the worst-case tornado found in Section 6.2 is used. The results across 10 replications are shown in Table 4, which shows that for both cases the robust optimal population dislocation is significantly less than the minimum of population dislocation with random retrofitting. As with the previous experiments, the greedy policy is a reasonable proxy for the optimal solution. The plot in Fig. 10 provides a visualization of the significant gap between robust optimal value (stars) and the distribution of values for random retrofitting plans. We can also observe that population dislocation reduces by increasing the amount of budget.

The results of this section show that the decision to use the proposed two-stage robust method yields significant reductions in population dislocation compared to random policies when evaluating in terms of worst-case outcomes. Remarkably, the greedy policy provides very good results, comparable to those of the optimal policy. However, in this context, any slight reduction in population dislocation implies significant societal effects, because even one prevented dislocation results in one less household not having to deal with profound disruptions in their personal lives, their families, and their community. Moreover, considering that the proposed approach provides optimality guarantees (whereas the greedy does not), we could confidently say that the community stakeholders would be taking the best decision possible under the assumptions provided.

Table 5

Comparison of population dislocation statistics for the worst-case tornado and simulated tornadoes by using the optimal retrofitting plan.

Budget	Population dislocation				
	Worst-case tornado	Average	Maximum	Minimum	Standard deviation
\$0M	16 318	10 976	16 318	1995	3039
\$15M	13 408	10 019	13 401	1995	2562
\$30M	13 091	9773	12 991	1881	2477

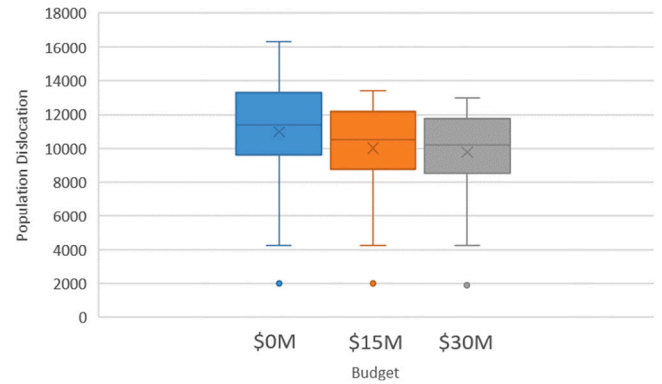


Fig. 11. Box and Whisker plots for Table 5 experiments. It shows the distribution of dislocation values for simulated tornadoes.

In the next section we complement these experiments by analyzing how the optimal policy behaves when random tornadoes, instead of the worst-case scenario, happen.

6.4. Robust strategies in random scenarios

In this section, we fix the optimal retrofitting plans and compare the robust population dislocation values resulting from having 50 simulated 5-miles tornadoes, see Table 5 and Fig. 11.

The results for \$0M in Table 5 and Fig. 11 show the difference between the worst-case tornado 16318 and the average value among simulation results which is 10976. As expected, by having more budget, the average population dislocations in the simulations reduce from 10976 to 10019 for \$15M and to 9773 for \$30M. In addition, the gap between the robust and average simulation values for \$15 and \$30M are smaller than the gap for the no-budget case.

These observations show the following about the optimal retrofitting policy: first, that there can be a large variability for population dislocation assuming any tornado is possible. Second, that the variability seems to decrease as more budget is available, and third, that the worst-case scenario is not that far away from simulated scenarios. This last point also suggests that the worst-case is not as rare as one might anticipate, at least in this specific instance.

6.5. Importance of the infeasible triple cuts

In this section we show the important effects that adding triple cuts has on the proposed DBC approach. Table 6 shows the results of running the DBC algorithm without the infeasible triple cuts for a tornado with a length of 5 miles assuming 100 locations. Specifically,

Table 6

Solving the two-stage robust optimization model for 5-mile tornado without infeasible triple cuts for 100 locations.

Length	Budget (\$)	Population dislocation	CCG run time (s.)	CCG iteration	DBC run time (s.)	Callback run time (s.)	$\tilde{U}^h$ feasibility run time (s.)	Budget proportion for retrofitting	Budget proportion for recovery
5	0 M	–	3602	1	3602	206	1	–	–
	15 M	–	3606	1	3605	369	1	–	–
	30 M	–	3603	1	3603	239	1	–	–

Table 7

Solving the two-stage robust optimization model with relaxed second stage problem, after 60 days of recovery in Joplin, MO.

# center locations	Length	Budget (\$)	Population dislocation	CCG run time (s.)	CCG iteration	DBC run time (s.)	Callback run time (s.)	$\tilde{U}^h$ feasibility run time (s.)	Budget proportion for retrofitting	Budget proportion for recovery
100	5	0 M	16318	57	2	57	5	1	–	–
		15 M	13333	159	3	159	10	4	0.6	0.4
		30 M	13050	166	3	166	8	2	0.3	0.7
	∞	0 M	17293	65	2	64	7	0	–	–
		15 M	14201	81	3	81	4	0	0.7	0.3
		30 M	13834	88	3	88	5	0	0.4	0.6
150	5	0 M	14850	763	2	763	8	1	–	–
		15 M	12252	2314	5	2313	18	6	0.8	0.2
		30 M	11906	3731	7	3729	32	10	0.4	0.6
	∞	0 M	15989	411	2	411	7	0	–	–
		15 M	13136	977	4	976	13	0	0.8	0.2
		30 M	12749	1067	4	1066	15	0	0.4	0.6
200	5	0 M	15002	3044	2	3044	30	4	–	–
		15 M	12711	4728(5%)	2	4728	39	10	0.6	0.4
		30 M	12541	4730(6%)	2	4730	20	4	0.3	0.7
	∞	0 M	15974	1570	2	1570	8	0	–	–
		15 M	13203	4472(2%)	3	4471	13	0	0.7	0.3
		30 M	12745	4313(1%)	4	4312	24	0	0.4	0.6

when computing $\Phi(f)$ using formulation (6), for a master feasible solution f we initialize C^0 only with the infeasible pair cuts; we do not use any infeasible triple cut.

As the results show, without any triple cut the problem is not solved within a time limit of 1 h for any value of budget. Since all the runs stopped at the first iteration of the algorithm while trying to solve the subproblem $\Phi(f)$, it is concluded that without infeasible triple cuts, the subproblem is unable to find a feasible tornado scenario to the master problem within the time limit. More specifically, if no infeasible triple cut is included in C^0 then no solution z that satisfies all of the constraints of problem (6) is found by the DBC algorithm within the time limit. Thus, initializing C^0 with at least some of the infeasible triple cuts is fundamental for the DBC algorithm to find feasible and optimal solutions.

6.6. Partial recovery

We close this section by considering experiments where we allow the second stage decision variables $r_{\ell,sp}$ to take continuous values between 0 and 1. This relaxation can be understood as a ‘partial recovery’ and implies that the decision-maker is allowed to cover only a fraction of the recovery costs. The results for 100, 150, and 200 location in Joplin with the relaxed second stage problem are summarized in Table 7.

As can be seen in Table 7, the relaxed variant results in slightly lower population dislocation values than the binary recovery results in Table 2, because there is more decision-making flexibility. The optimal solutions are slightly different with respect to the binary variant; see Fig. 12 that shows the solution for the 5-mile tornado over 100 locations. The optimal recoveries in the fractional model are: with a budget of \$15M only 82% of the recovery costs are covered for the location in green in Fig. 12(b). With a budget of \$30M, the recovery costs covered are 100%, 19%, 100%, and 34%, from left to right in Fig. 12(d), respectively.

We also observe that the run times are almost the same for both variants, which is somewhat surprising because, in theory, the relaxed problem is easier to solve than the binary problem. In practice, however, solving the second-stage problem in both variants can be done quickly. An alternative approach worth noting for tackling the subproblem $\phi(f)$ in (5) for the relaxed variant is to employ the dual linear program derived from the relaxation of $Q(z, f)$ in (2). Then, the subproblem $\phi(f)$ can be reformulated to a single-level mixed-integer problem that directly determines the worst tornado scenario for a given retrofitting strategy f . In our approach, this would imply not having to decompose over the r variables when solving the subproblem, which can potentially improve computational times.

7. Conclusions

Motivated by the need of data-driven methods to allocate resources in retrofitting for tornado-prone areas, we consider a two-stage robust optimization model to determine retrofitting and recovery strategies before and after the realization of an uncertain tornado disaster. We assume that the decision-maker (a combination of public–private agencies) has to allocate a budget in retrofitting and recovery actions, to minimize population dislocation. We propose a two-stage optimization model where the first-stage variables determine the retrofitting strategies and the second-stage variables determine the recovery strategy, assuming that for any retrofitting action, the corresponding worst-case tornado happens. Given that the uncertainty corresponds to the possible tornado paths, the resulting problem is a two-stage mixed-integer robust optimization problem with a mixed-integer non-linear uncertainty set.

We show that the proposed problem is NP-hard. A standard approach to solve the problem is a column-and-constraint generation algorithm that needs to solve a max–min subproblem at each iteration. We embed a decomposition branch-and-cut algorithm in the column-and-constraint solution method to address the non-convex structure of the bilevel subproblem. We propose two sets of conflict constraints

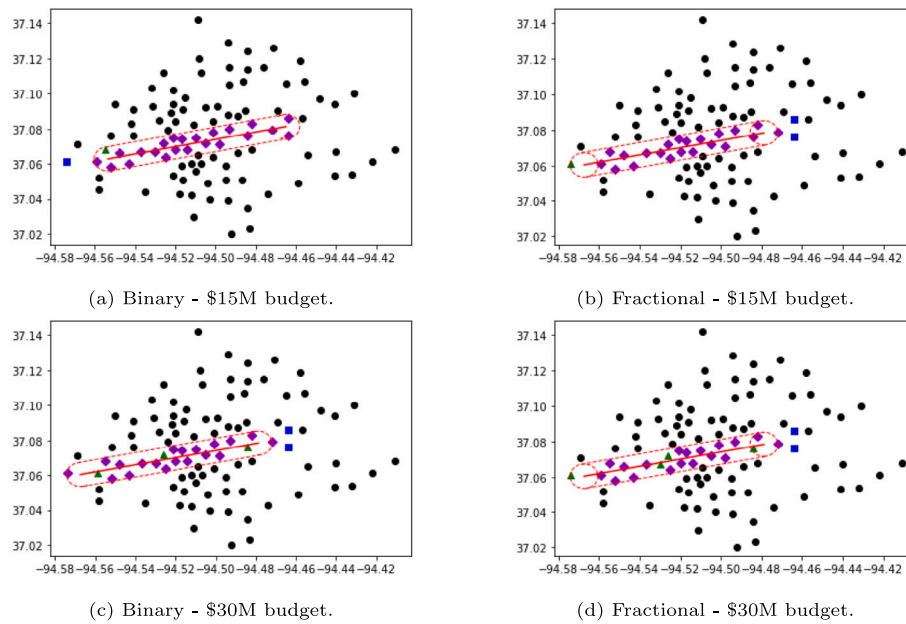


Fig. 12. Comparison between binary versus relaxed second stage problem for 100 locations with \$15M and \$30M budget.

and a custom separation procedure to implement the decomposition branch-and-cut algorithm. Using data from INCORE, we use our model to provide the optimal retrofitting and recovery strategies for the city of Joplin, Missouri against the worst-case scenario. The results show that the optimal policy exhibits a diminishing returns behavior that sets-in quickly; prioritizes the high-density city center; and prefers to retrofit and recover than to only retrofit. The proposed policy also outperforms other benchmark.

Future work will focus on modeling tornado damages assuming that there might be several levels of disruptions. Additionally, future research would analyze other social vulnerability indices to measure community resilience under the risk of a tornado. One line of research might study the population dislocation in residential areas with different income levels after certain days of a tornado occurrence, in order to recognize more vulnerable locations that require immediate agencies' attention. Another line of research might take our proposed framework to study other natural disasters like hurricanes with minor modifications in the uncertainty model and the set of retrofitting strategies.

CRedit authorship contribution statement

Mehdi Ansari: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Juan S. Borrero:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Andrés D. González:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.ejor.2025.03.001>.

References

- An, Y., & Zeng, B. (2015). Exploring the modeling capacity of two-stage robust optimization: Variants of robust unit commitment model. *IEEE Transactions on Power Systems*, 30(1), 109–122. <http://dx.doi.org/10.1109/TPWRS.2014.2320880>.
- An, Y., Zeng, B., Zhang, Y., & Zhao, L. (2014). Reliable p-median facility location problem: two-stage robust models and algorithms. *Transportation Research Part B: Methodological*, 64, 54–72.
- Ansari, M., Borrero, J. S., & Lozano, L. (2022). Robust minimum-cost flow problems under multiple ripple effect disruptions. *INFORMS Journal on Computing*.
- Atamtürk, A., & Zhang, M. (2007). Two-stage robust network flow and design under demand uncertainty. *Operations Research*, 55(4), 662–673.
- Ben-Tal, A., Do Chung, B., Mandala, S. R., & Yao, T. (2011). Robust optimization for emergency logistics planning: Risk mitigation in humanitarian relief supply chains. *Transportation Research Part B: Methodological*, 45(8), 1177–1189.
- Ben-Tal, A., Goryashko, A., Guslitzer, E., & Nemirovski, A. (2004). Adjustable robust solutions of uncertain linear programs. *Mathematical Programming*, 99(2), 351–376.
- Ben-Tal, A., Hazan, E., Koren, T., & Mannor, S. (2015). Oracle-based robust optimization via online learning. *Operations Research*, 63(3), 628–638.
- Ben-Tal, A., & Nemirovski, A. (1999). Robust solutions to uncertain linear programs. *Operations Research Letters*, 25, 1–13.
- Berkes, F., & Ross, H. (2013). Community resilience: Toward an integrated approach. *Society & Natural Resources*, 26, 5–20. <http://dx.doi.org/10.1080/08941920.2012.736605>.
- Bertsimas, D., & Sim, M. (2004). The price of robustness. *Operations Research*, 52(1), 35–53.
- Borrero, J. S., & Lozano, L. (2021). Modeling defender-attacker problems as robust linear programs with mixed-integer uncertainty sets. *INFORMS Journal on Computing*, 33(4), 1570–1589.
- Che, A., Li, J., Chu, F., & Chu, C. (2024). Optimizing emergency supply pre-positioning for disaster relief: A two-stage distributionally robust approach. *Computers & Operations Research*, 166, Article 106607.

- Cheng, C., Adulyasak, Y., & Rousseau, L.-M. (2021). Robust facility location under demand uncertainty and facility disruptions. *Omega*, 103, Article 102429.
- Cheng, C., Yu, Q., Adulyasak, Y., & Rousseau, L.-M. (2024). Distributionally robust facility location with uncertain facility capacity and customer demand. *Omega*, 122, Article 102959.
- Dang, D.-C., Currie, C. S., Onggo, B. S., Chaerani, D., & Achmad, A. L. H. (2023). Budget allocation of food procurement for natural disaster response. *European Journal of Operational Research*, 311(2), 754–768.
- Ding, T., Li, C., Yang, Y., Jiang, J., Bie, Z., & Blaabjerg, F. (2017). A two-stage robust optimization for centralized-optimal dispatch of photovoltaic inverters in active distribution networks. *IEEE Transactions on Sustainable Energy*, 8(2), 744–754. <http://dx.doi.org/10.1109/TSTE.2016.2605926>.
- Dumouchelle, J., Julien, E., Kurtz, J., & Khalil, E. B. (2024). Neur2RO: Neural two-stage robust optimization. In *The twelfth international conference on learning representations*. URL <https://openreview.net/forum?id=T5Xb0iGCCv>.
- Duran-Mateluna, C., Ales, Z., Elloumi, S., & Jorquera-Bravo, N. (2023). Robust MILP formulations for the two-stage weighted vertex p-center problem. *Computers & Operations Research*, 159, Article 106334.
- Fan, F., & Pang, W. (2019). Stochastic track model for tornado risk assessment in the US. *Frontiers in Built Environment*, 5, 37.
- Farokhnia, K., van de Lindt, J. W., & Koliou, M. (2020). Selection of residential building design requirements to achieve community functionality goals under tornado loading. *Practice Periodical on Structural Design and Construction*, 25(1), Article 04019035.
- FEMA (2021). Natural hazard retrofit program toolkit. https://www.fema.gov/sites/default/files/documents/fema_natural-hazards-retrofit-program-toolkit.pdf. (Accessed January 2023).
- FEMA (2022a). Resources for repairing, retrofitting and rebuilding after a tornado. <https://www.fema.gov/fact-sheet/resources-repairing-retrofitting-and-rebuilding-after-tornado-0>. (Accessed January 2023).
- FEMA (2022b). Tornado-alerts and warnings. Online at <https://community.fema.gov/ProtectiveActions>. (Accessed November 2022).
- Gabrel, V., Lacroix, M., Murat, C., & Remli, N. (2014). Robust location transportation problems under uncertain demands. *Discrete Applied Mathematics*, 164, 100–111.
- Goerigk, M., & Kurtz, J. (2025). Data-driven prediction of relevant scenarios for robust combinatorial optimization. *Computers & Operations Research*, 174, Article 106886.
- Ho-Nguyen, N., & Kilinç-Karzan, F. (2018). Online first-order framework for robust convex optimization. *Operations Research*, 66(6), 1670–1692.
- IN-CORE (2022). Interdependent networked community resilience modeling environment. <https://incore.ncsa.illinois.edu/>. (Accessed October 2022).
- Jabr, R. A., Džafić, I., & Pal, B. C. (2015). Robust optimization of storage investment on transmission networks. *IEEE Transactions on Power Systems*, 30(1), 531–539. <http://dx.doi.org/10.1109/TPWRS.2014.2326557>.
- Jiang, R., Zhang, M., Li, G., & Guan, Y. (2012). Benders' decomposition for the two-stage security constrained robust unit commitment problem. In *IIE annual conference proceedings* (p. 1). Institute of Industrial and Systems Engineers (IISE).
- Koliou, M., & van de Lindt, J. W. (2020). Development of building restoration functions for use in community recovery planning to tornadoes. *Natural Hazards Review*, 21(2), Article 04020004.
- Koliou, M., van de Lindt, J. W., McAllister, T. P., Ellingwood, B. R., Dillard, M., & Cutler, H. (2020). State of the research in community resilience: progress and challenges. *Sustainable and Resilient Infrastructure*, 5, 131–151. <http://dx.doi.org/10.1080/23789689.2017.1418547>.
- Kuligowski, E. D., Lombardo, F. T., Phan, L., Levitan, M. L., & Jorgensen, D. P. (2014). Final report, national institute of standards and technology (NIST) technical investigation of the may 22, 2011, tornado in joplin, missouri.
- Laporte, G., & Louveaux, F. V. (1993). The integer L-shaped method for stochastic integer programs with complete recourse. *Operations Research Letters*, 13, 133–142.
- Li, Y., Zhang, F., Li, Y., & Wang, Y. (2021). An improved two-stage robust optimization model for CCHP-P2G microgrid system considering multi-energy operation under wind power outputs uncertainties. *Energy*, 223, Article 120048.
- Lin, X., Janak, S. L., & Floudas, C. A. (2004). A new robust optimization approach for scheduling under uncertainty: I. bounded uncertainty. *Computers & Chemical Engineering*, 28(6–7), 1069–1085.
- Ma, S., Chen, B., & Wang, Z. (2018). Resilience enhancement strategy for distribution systems under extreme weather events. *IEEE Transactions on Smart Grid*, 9(2), 1442–1451.
- MacQueen, J. (1967). Classification and analysis of multivariate observations. In *5th berkeley symp. math. statist. probability* (pp. 281–297).
- Masoomi, H., Ameri, M. R., & van de Lindt, J. W. (2018). Wind performance enhancement strategies for residential wood-frame buildings. *Journal of Performance of Constructed Facilities*, 32(3), Article 04018024.
- Masoomi, H., & van de Lindt, J. W. (2018). Restoration and functionality assessment of a community subjected to tornado hazard. *Structure and Infrastructure Engineering*, 14(3), 275–291.
- Matthews, L. R., Gounaris, C. E., & Kevrekidis, I. G. (2019). Designing networks with resiliency to edge failures using two-stage robust optimization. *European Journal of Operational Research*, 279(3), 704–720.
- Mutapcic, A., & Boyd, S. (2009). Cutting-set methods for robust convex optimization with pessimizing oracles. *Optimization Methods & Software*, 24(3), 381–406.
- Neyshabouri, S., & Berg, B. P. (2017). Two-stage robust optimization approach to elective surgery and downstream capacity planning. *European Journal of Operational Research*, 260(1), 21–40.
- NOAA (2022a). National centers for environmental information, tornadoes statistics. Online at <https://www.ncei.noaa.gov>. (Accessed September 2022).
- NOAA (2022b). National weather service, the enhanced fujita scale (EF scale). <https://www.weather.gov>. (Accessed September 2022).
- NOAA (2022c). Storm predication center - national weather service. Online at <https://www.spc.noaa.gov/>. (Accessed October 2022).
- NOAA (2022d). Supercells. Online at <https://www.spc.noaa.gov/misc/AbtDerechos/supercells.htm>. (Accessed November 2022).
- NOAA (2022e). Tornadoes frequently asked questions. Online at <https://www.weather.gov/lmk/tornadoesfaq>. (Accessed November 2022).
- NOAA (2022f). Weather related fatality and injury statistics. Online at <https://www.weather.gov/hazstat/>. (Accessed November 2022).
- NOAA (2023). Damage assessment toolkit. Online at <https://apps.dat.noaa.gov/stormdamage/damageviewer/>. (Accessed March 2023).
- NWC (2023). Historical data of oklahoma county tornadoes. Online at <https://www.weather.gov/oun/tornadodata-county-ok-oklahoma>. (Accessed January 2023).
- Ripberger, J. T., Jenkins-Smith, H. C., Silva, C. L., Czajkowski, J., Kunreuther, H., & Simmons, K. M. (2018). Tornado damage mitigation: Homeowner support for enhanced building codes in oklahoma. *Risk Analysis*, 38(11), 2300–2317.
- Shams, M. H., Shahabi, M., MansourLakouraj, M., Shafie-khah, M., & Catalão, J. P. (2021). Adjustable robust optimization approach for two-stage operation of energy hub-based microgrids. *Energy*, 222, Article 119894.
- Simmons, K. M., Kovacs, P., & Kopp, G. A. (2015). Tornado damage mitigation: Benefit-cost analysis of enhanced building codes in oklahoma. *Weather, Climate, and Society*, 7(2), 169–178.
- Standohar-Alfano, C. D., van de Lindt, J. W., & Ellingwood, B. R. (2017). Vertical load path failure risk analysis of residential wood-frame construction in tornadoes. *Journal of Structural Engineering*, 143(7), Article 04017045.
- Stoner, M., & Pang, W. (2021). Tornado hazard assessment of residential structures built using cross-laminated timber and light-frame wood construction in the US. *Natural Hazards Review*, 22(4), Article 04021032.
- Strader, S. M., Pingel, T. J., & Ashley, W. S. (2016). A Monte Carlo model for estimating tornado impacts. *Meteorological Applications*, 23(2), 269–281.
- Subramanyam, A. (2022). A Lagrangian dual method for two-stage robust optimization with binary uncertainties. *Optimization and Engineering*, 23(4), 1831–1871.
- Takeda, A., Taguchi, S., & Tütüncü, R. (2008). Adjustable robust optimization models for a nonlinear two-period system. *Journal of Optimization Theory and Applications*, 136(2), 275–295.
- The White House (2022). FACT SHEET: Biden–Harris administration launches initiative to modernize building codes, improve climate resilience, and reduce energy costs. <https://www.whitehouse.gov/briefing-room/statements-releases/2022/06/01/fact-sheet-biden-harris-administration-launches-initiative-to-modernize-building-codes-improve-climate-resilience-and-reduce-energy-costs/>. (Accessed: January 2023).
- Thiele, A., Terry, T., & Epelman, M. (2009). Robust linear optimization with recourse. *Rapport Technique*, 4–37.
- Tsang, M. Y., Shehadeh, K. S., & Curtis, F. E. (2023). An inexact column-and-constraint generation method to solve two-stage robust optimization problems. *Operations Research Letters*, 51(1), 92–98.
- van de Lindt, J. W., & Dao, T. N. (2009). Performance-based wind engineering for wood-frame buildings. *Journal of Structural Engineering*, 135(2), 169–177.
- Velasquez, G. A., Mayorga, M. E., & Özalpın, O. Y. (2020). Prepositioning disaster relief supplies using robust optimization. *IIE Transactions*, 52(10), 1122–1140.
- Wang, J., Cao, S., Pang, W., & Cao, J. (2017). Experimental study on effects of ground roughness on flow characteristics of tornado-like vortices. *Boundary-Layer Meteorology*, 162, 319–339.
- Wang, C., Miao, L., Zhang, C., Wu, T., & Liang, Z. (2024). Robust optimization for the integrated berth allocation and quay crane assignment problem. *Naval Research Logistics*, 71(3), 452–476.
- Wang, X. J., & Paul, J. A. (2020). Robust optimization for hurricane preparedness. *International Journal of Production Economics*, 221, Article 107464. <http://dx.doi.org/10.1016/j.ijpe.2019.07.037>.
- Wang, W., Van De Lindt, J. W., Rosenheim, N., Cutler, H., Hartman, B., Sung Lee, J., & Calderon, D. (2021). Effect of residential building wind retrofits on social and economic community-level resilience metrics. *Journal of Infrastructure Systems*, 27(4), Article 04021034.
- Wang, C., & Zhong, L.-M.-T. (2024). Reliable design of humanitarian supply chain under correlated disruptions: a two-stage distributionally robust approach. *Annals of Operations Research*, 1–49.
- Wen, Y. (2021). *Development of multi-objective optimization model of community resilience on mitigation planning* (Ph.D. thesis), University of Oklahoma.
- Xiong, P., Jirutitijaroen, P., & Singh, C. (2017). A distributionally robust optimization model for unit commitment considering uncertain wind power generation. *IEEE Transactions on Power Systems*, 32(1), 39–49.
- Yao, T., Mandala, S. R., & Do Chung, B. (2009). Evacuation transportation planning under uncertainty: a robust optimization approach. *Networks and Spatial Economics*, 9(2), 171.

- Yin, Y., Xu, X., Wang, D., Yu, Y., & Cheng, T. (2024). Two-stage recoverable robust optimization for an integrated location-allocation and evacuation planning problem. *Transportation Research Part B: Methodological*, 182, Article 102906.
- Yuan, W., Wang, J., Qiu, F., Chen, C., Kang, C., & Zeng, B. (2016a). Robust optimization-based resilient distribution network planning against natural disasters. *IEEE Transactions on Smart Grid*, 7, 2817–2826. <http://dx.doi.org/10.1109/TSG.2015.2513048>.
- Yuan, W., Wang, J., Qiu, F., Chen, C., Kang, C., & Zeng, B. (2016b). Robust optimization-based resilient distribution network planning against natural disasters. *IEEE Transactions on Smart Grid*, 7(6), 2817–2826.
- Zeng, B., & Zhao, L. (2013). Solving two-stage robust optimization problems using a column-and-constraint generation method. *Operations Research Letters*, 41(5), 457–461.
- Zhao, L., & Zeng, B. (2012). Robust unit commitment problem with demand response and wind energy. In *2012 IEEE power and energy society general meeting* (pp. 1–8). IEEE.